

LOESCHE-Mill LM 46.2+2 CS**Table of Contents****Designation****BA Number**

Safety

BA 911092-00-4

Structure and Function

BA 911334-12-4

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1. General

These operating instructions LOESCHE Mill - "Safety" shall be considered as a supplement to the Roller Grinding Mill - "Safety" instructions. It must be ensured that these operating instructions will be studied in detail prior to operation of the mill.

2. Notes on Safety at Work

Particular attention shall be paid to the following notes as to safety at work:

- During grinding, no persons shall be allowed to stay within range of mill column ("LM base" assembly). Accident hazard!
- The operator shall make sure that operation of the hydraulic accumulators of hydropneumatic roller cushioning system will be in compliance with all statutory regulations applicable in the designated country.
- Maintenance work on the hydraulic system shall not be performed unless the system is depressurized and has been emptied completely.
- Only nitrogen shall be used for refilling of the hydraulic accumulators. Explosions may be caused by the presence of oxygen or air in connection with oil.
- Only the original filling equipment shall be used for filling of hydraulic accumulators. Attention should be paid to the instructions contained in the hydraulic diagram.
- Quality, cleanliness and operating viscosity of the hydraulic medium are decisive for safety of operation, economy and service life of the plant.
- Prior to entering the grinding area, there must be made sure that the mill motor cannot be started.
- While the rollers are being swung out of and into position, persons shall be allowed to stay neither within the grinding area nor within range of the rollers. Accident hazard!
- In case of maintenance work necessitating dismantling of the platform safety belts shall be used, in order to prevent accidents by falling.
- All statutory accident prevention regulations in the designated country shall be observed when performing any maintenance work and inspections.

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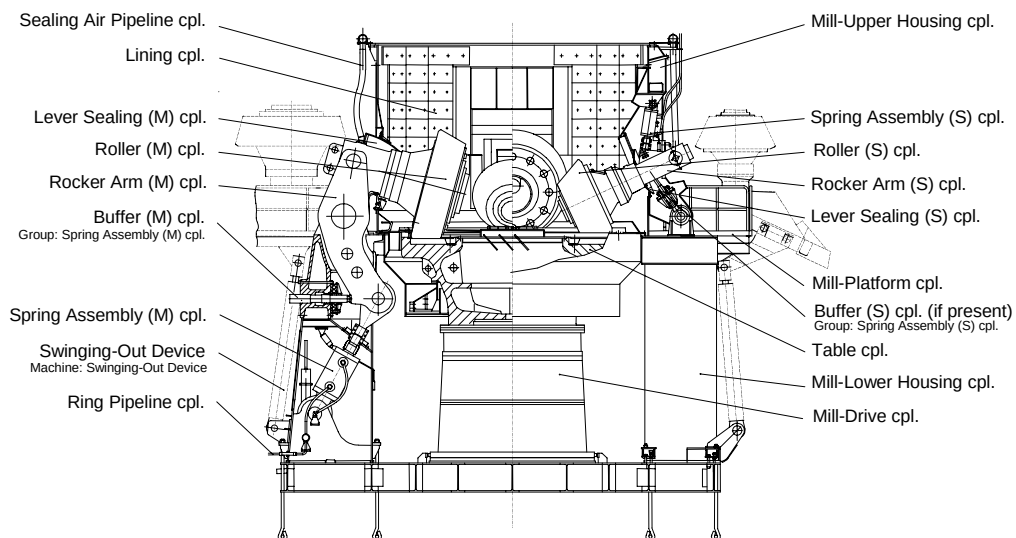
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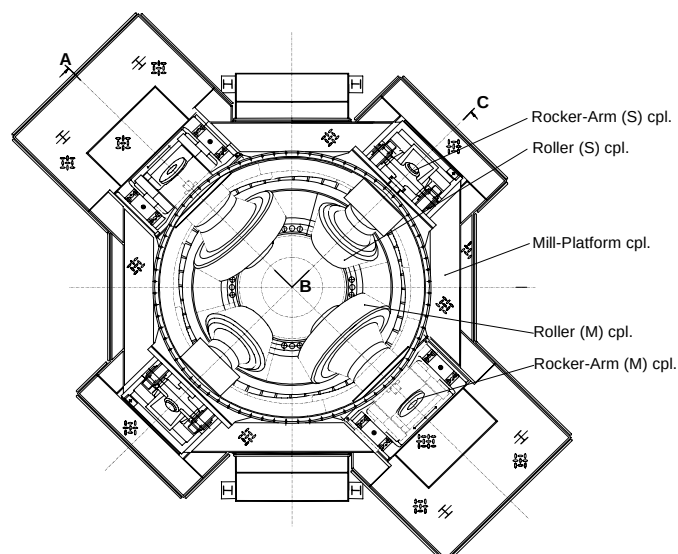
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1. Structure

Section A-B-C



principle sketches



2. Function

The mill is a negative pressure, air-swept machine for grinding and drying the feed.

The feed which lies upon the table and is crushed by the rollers, consists of a small proportion of new feedstock with a high moisture content and a large proportion of partially ground material with a low moisture content, which has been rejected as being oversize, by the classifier. Both materials are highly aerated as they arrive on the table.

The feed is directed by centrifugal force under the S-rollers which prepare it by de-aerating and partially compressing the loose particles together. The S-rollers are held at a distance above the grinding plate in their lightweight rocker arms.

The M-rollers, mounted in massive rocker arms, grind the prepared feedstock largely by compression. The force is created by the weight of the roller and an additional force generated by the hydraulic system.

The rollers are rotated by friction with the grinding bed. Sensors on the rocker arm and stop-buffers (spring-loaded travel limiters) prevent any metallic contact between the rollers and the table.

The crushed grit is thrown outwards by centrifugal force and flows over the rim of the table. Here it is caught in the gas stream which flows upwards through the louvre ring (see assembly "Table cpl."). The dust/gas mixture flows through the grinding chamber, which is enclosed by the mill upper housing with its special armour lining, to the classifier. This is mounted or welded to the flange of the mill upper housing.

The mill can be started in filled condition (after a short stop) by the mill motor without the help of an auxiliary drive. To off-load the motor, the rollers are hydraulically lifted. This is achieved by applying pressurised oil to the piston-side of the spring assembly hydraulic cylinders.

Mill-Lower-Housing cpl.

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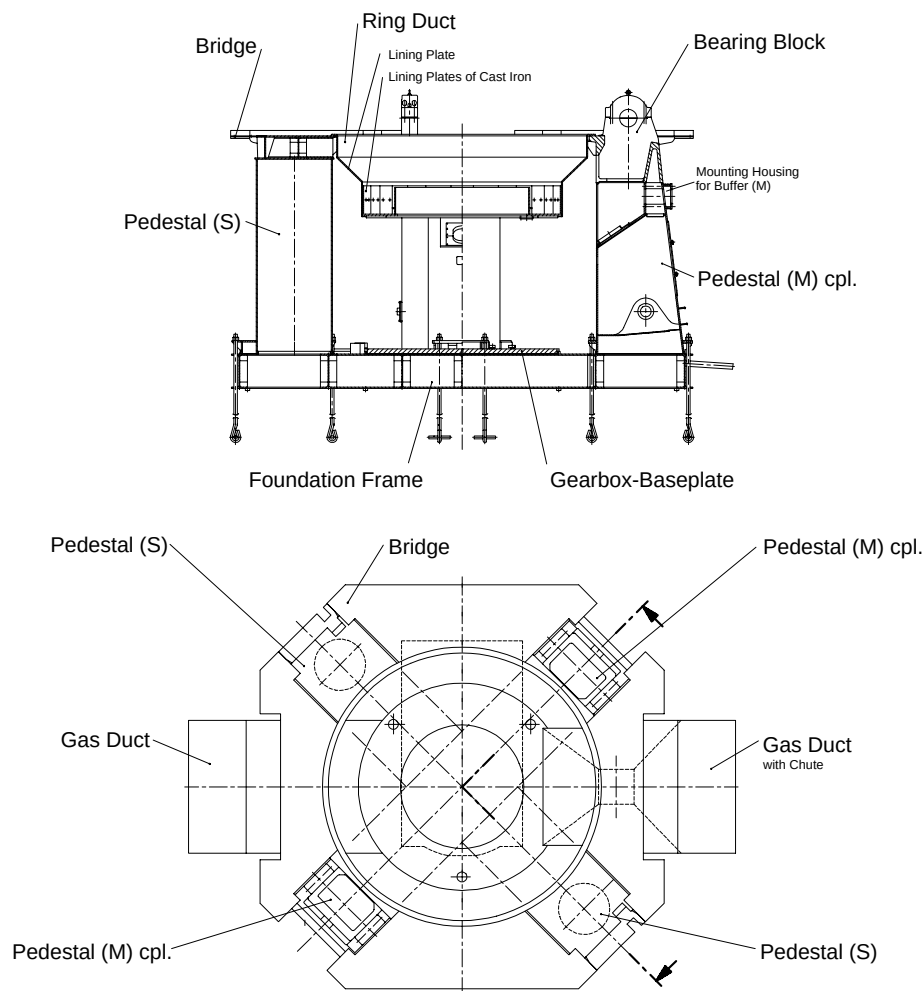
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1. Structure



Principle Sketches

The group "Mill-Lower-Housing cpl." consists primarily of the foundation frame, the base plate, the pedestals, the bridges, the ring duct and the gas duct with integrated (reject) chute.

The Mill-Lower-Housing is a combined cast/welded construction whose foundation frame is held in the foundation by anchor bolts and by these bound to the reinforcement in the cast concrete.

The various components of the group are described in detail as follows:

- **Foundation Frame**

The foundation frame is a weldment made and serves as a support for the pedestals and the mill gearbox via the gearbox base plate. It is divided into a main section and subsections. This is necessary for transport purposes. These construction elements are welded into one unit at site. To help correct assembly, corresponding fixing points for the separate parts are attached by the manufacturer.

- **Gearbox Base Plate**

The base plate for the gearbox is a machined plate. It supports the mill gearbox and connects it with the Mill-lower housing. The plate is held in place through application of a resin compound and finally pinned in its definitive position. After proper alignment by means of jacking screws to achieve a level attitude of the plate, bolts are used to connect the plate to the foundation frame.

The pins between the base plate and the frame have the purpose of absorbing shear forces generated during the installation and extraction of the gearbox, that is to say, when the gearbox is drawn across the base plate the plate is held in place by the pins. Consequently, the resin layer is not subjected to shear forces but only compression forces.

Whilst the mill is in operation, the bottom flange of the gearbox, the gearbox base plate and the foundation frame are held rigidly in place by means of bolts. These bolts are relieved by wedges which support the gearbox on the main frame whilst the mill is in operation. The bottom flange of the gearbox has been provided with machined stopper surfaces. Back torque of the gearbox is, consequently, transmitted directly to the foundation frame without stressing the screws and the resin bound between the base plate and the foundation frame.

- **M-Pedestal cpl.**

The both M-pedestals of the Mill-lower housing are standing opposite to each other. They are combined cast steel/welded construction. Onto a sheet metal box plinth a steel cast bearing block is welded.

The bearing housing in the bearing block is manufactured in two parts. The bearing covers are bolted to the bearing block with pins and screws. One of each pair of bearing housings is provided with tapped holes to receive the bolts for the axial thrust discs of the rocker arm axle.

Plates with cut-outs are welded into the pedestal and serve as diagonal braces to increase the rigidity of the pedestal against torsion.

A housing for the buffer of the M-rocker arm is welded into the M-pedestal below the bearing block. It transfers the loads applied to the buffer into the steelwork of the pedestal.

In the pedestal base straps are welded and bored through for the location of the spring pin for the M-Spring assembly. Here the hydraulic cylinder with its bottom eye is suspended.

A door attached to the pedestal closes the mounting entrance for the hydraulic cylinder of the M-Spring assembly. The door is equipped with a special eccentric grip for quick locking.

Covers on the side wall of the pedestal closes the entry for the hydraulic lines between the hydraulic cylinder located inside the pedestal and the hydraulic accumulators outside. At the same time, they support the hydraulic pipes (see assembly "M-Spring Assembly cpl.").

S-Pedestal

The two S-pedestals stand opposite to one another and at 90° to the M-pedestals.

The S-pedestals are fabricated as a rolled and welded column. At the upper end a box construction is welded with the bridges, ring duct and mill body. The column is supported by the foundation frame.

The upper portion of the S-pedestal has a flat surface. At site assembly the S-buffer for the S-rocker arm is aligned and welded to this support.

Bridges

The bridges are box girder constructions. They are inserted, at the height of the M-pedestal bearing blocks between two corresponding M+S pedestals, bolted and then welded to the M-bearing block and the upper portion of the S-pedestal to form a torsionally stiff ring connection.

Ring Duct

The ring duct surrounds the table. It serves to direct the hot gas to the louvre ring (see construction group "Table cpl.").

The ring duct is a weldment. It is connected to the M-and S-stands and bridges by a flange. It is made air-tight by welding.

The bottom of the ring duct is double-walled. This box-type construction achieves a high degree of stiffness as well as providing thermal insulation between the hot gas and the gearbox below.

On the underside of the ring duct there is a multi-part ring. This serves as a housing for an air-seal to the throat of the table.

The ring duct is protected by special measures in these areas, where it will suffer from higher wear. Therefore the conical upper section is provided with segments of high-abrasion resistant sheet-steel and the cylindrical lower section with ribbed lining plates made of cast iron. These lining plates are bolted on and their condition can be checked through inspection doors and if necessary changed. The bottom of the ring duct will be protected by a layer of reject material which collects here.

Gas Duct

The gas ducts may be considered as a continuation of the hot gas pipeline to the mill and transfer the gas to the ring duct. One gas duct is provided in its lower section with a chute through which the reject material falls.

The base of the gasduct lies at a higher level than the base of the ringduct so that the scrappers cannot throw rejected material into the gasduct inlet.

Each gas duct is welded on site to an air-tight unit with the ring duct.

2. Function

The Mill-lower housing has two functions. It serves as a base and consequently as a support for the grinding mechanism and the power transmission components, and also to transfer hot gases to the louvre ring surrounding the table. The louvre ring is located in the upper portion of the ring duct.

The grinding forces are transmitted from the M-rollers via the table and the mill gearbox to the foundation frame and from here into the foundation.

The path of the forces from the foundation frame is over the linkage (in the lower section of the pedestals) in which the hydraulic cylinders are located, through the hydraulic cylinders. From here they run through the M-rocker arms in the bearing blocks, over the M-rollers onto the table and through the gearbox into the foundation frame where it is closed.

The table exerts by its peripheral force during grinding a bending and torsional force on every M-rocker arm and bearing block. Those forces which are directed downward are led by the M-pedestals into the foundation. Here they are finally absorbed in the foundation. The torsional forces resultant from the grinding process and are led into the bearing blocks which are integrated in a torsional rigid unit that is consisting of the bearing blocks of the pedestals and the bridges that are situated in between the M-pedestals.

The hot air guiding parts, gas canal/ring canal, are only connected to the load directing parts by a flange. In this way heat expansion in the hot gas guiding parts cannot exert additional forces into the load guiding parts of the pedestal.

The hot gas canals conduct the primary air from the air heater into the Mill-lower housing.

The cross-section of the gas channels is dimensioned so that the flow resistance is as small as possible in all operating states of the mill. The hot gases are led around the table collar by the ring channel. The gas channel is optimal developed in its volume so that the velocity energy of the hot gases is reduced for the most part and changed in static pressure. This leads to an equal gas distribution in the area of the grinding table collar and the gases can flow into the grinding chamber through the louvre ring without considerable streamer formation.

Foreign bodies which are discharged by the mill through the louvre ring as well as grinding material which falls through the louvre ring in the Mill-lower housing in case of power failure are thrown into the inlet stud by scrapers arranged below the table. From here both glide to the reject outlet via the sloping bases of the gas channel.

3. Servicing

- Immediately after start-up tighten bolts on the covers. Repeat this operation after eight operating hours and finally secure heads as indicated in the drawing.
- Hexagonal nuts inside and outside the pedestal must be tightened and after approximately 100 hours of mill operation, the nuts are to be welded in accordance with instructions given on the assembly drawing.
- Hexagonal head bolts of the gearbox foot-flange must be tightened and secured as indicated in the gearbox-documentation.
- Wedges must be inspected after appr. 24 hours of operation and tightened if necessary. Wedges are then tack-welded lightly.

Mill Upper Housing cpl.

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1. Structure

The LM upper housing is of welded construction. The cylindrical housing will be welded to the LM lower housing. If required, the housing may be split for transportation. (The housing halves are provided with weld preparations at their joints. At the time of installation, the two halves held dimensionally stable by means of welded struts, will be welded together.)

The LM upper housing is provided with large-size openings for swivelling the M- and S-rocker arms complete with their associated rollers into and out of position. The openings will be closed by means of covers bolted to the housing. The openings for the rocker arms in the LM upper housing will be provided with appropriate sealing systems (please see assemblies "Rocker Arm: Sealing (M) cpl." and "Rocker Arm: Sealing (S) cpl.").

- M-Roller Covers

The M-roller covers are a plate/welded construction and bolted to corresponding flanged frames on the LM upper housing. The M-rocker arm collar will enter the grinding zone through an opening in the cover pocket. The rocker arm lead-in will be sealed by means of the "Rocker Arm Sealing (M) cpl." assembly (see Operating Instructions "M-Rocker Arm: Sealing cpl.").

- S-Roller Cover

The S-roller cover serves for closing an opening in the LM upper housing of the same size as the M-roller cover. Unlike the latter, however, the S-roller cover is braced with a welded-on box girder frame. In the upper part, a hollow profile is supported on a corresponding profile of the LM upper housing. This profile is provided with hinged brackets for accommodation of a hydraulic cylinder (assembly "Spring Assembly (S) cpl.") for the S-roller. Foot plates for accommodating plummer blocks for supporting the S-rocker arm assemblies are welded to both vertical box girders.

At their lower end, both box girders are extended to form hinged brackets with integrated joints. These are composed of hollow journals guided in bearing blocks.

At the time of erection at the final place of installation, the bearing blocks will be welded to the upper end plate of S-column (assembly "Mill Lower Housing cpl."). During erection at the site, shims will be used for accurate alignment of the bearing points and for welding without any distortion.

- **Mill Door**

Two doors with quick-release covers are provided for inspection of the mill interior.

2. Function

The LM upper housing serves the following tasks:

- formation of a grinding chamber retaining and guiding the dust/gas mixture between the table and the classifier;
- sealing of rocker arm lead-ins through the housing;
- supporting the forces from the S-roller systems via the integrated S-covers;
- supporting of classifier bolted to the flange of LM upper housing.

3. Maintenance

- The zero position of M-roller may be determined by way of a scale on the M-rocker arm and a pointer on the cover of LM upper housing. Herein, it shall be a precondition that the M-roller is lowered onto the table and that the pointer is set to "0" on the scale in this position. During operation of the mill, the scale will move along the stationary pointer. It will be possible then to obtain readings both of the mean grinding bed thickness and of thickness variations.
- Check bolts of S-cover for proper torquing at regular intervals and retighten, if necessary.

Mill-Upper Housing: Lining cpl.

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1. Structure

1.1 Housing Lining

The LM-housing is protected against wear by means of a cast lining. The castings are made ribbed plates, are bolted against the LM-upper housing and have a 2-point or 3-point support. Screws with a keyed head guarantee a tight fit until the liner plates are completely worn. The screws are tightened by standard nuts. The sealing of the thread shaft is achieved by means of a plastic ring with a steel cage against the LM-housing (see design drawing "Lining cpl.").

In order to accommodate unevenness of the LM-housing and to guarantee support without rocking, the rear surfaces of liner plates carry bosses in the area of the screw holes. Thus the 3-point (2-point) support is ensured.

Less endangered areas are protected by means of high wear-resistance plates, which are tack-welded to the LM-housing.

1.2 Cover Lining

The inside of the covers is protected by means of high wear-resistance plates, which are partly welded to the cover.

2. Function

The ribbed castings serve as "autogenous liners". This expression is used because they are self-protecting. The ribs are inclined rectangularly to the upwards directed P. A. flow. Thus grinded material fill the gaps between the ribs so that mainly dust slides over dust.

A part of the friction wear is converted into impact wear which leads to less erosion.

3. Servicing

- Tighten nuts of the special taper bolts following the sectional view.
- When replacing liner plates new bolts and sealing rings must be used.
- Worn liners must be replaced only after the ribs are washed off and the basic material starts eroding.
- Liners must not be replaced all at the same time. A partial replacement dependant on the wear condition is sufficient.
- Sheet plates should be replaced before showing holes.

Table cpl.

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1. Structure and Function

The functional parts of the assembly "Table cpl." are basically the table body, the grinding plate, the clamping ring, the dam ring, the louvre ring with armoured ring and the scrapers.

The primary function of the table is to support the grinding bed and transmit the force generated by the rollers into the gearbox below. It also transmits the torque of the gearbox into the grinding zone. Together with the ring duct (mill lower housing cpl.) the table creates a circular chamber out of which the primary gas flows over the louvre ring into the grinding chamber.

As one of the two grinding tools of the mill - the second tool are the rollers - the table is of such importance in view of the various spare part in the assembly which is described in more detail below:

- Table Body

The table is a heavy cast rotary body rigidly connected to the output flange of mill gearbox by means of bolts. The torque of the mill gear box is taken up by the table body by the friction force produced by the hexagon bolts. Additionally two drive studs are provided.

The table body has a horizontal surface, armour plated with the grinding plate which serves as the grinding path. Holes for bleeding off undesired iron parts run through grinding plate and table body.

The grinding forces generated by the rollers are passed through the table to the gearbox. Due to the mass of the table itself and the chosen material of manufacture, a proportion of the dynamic forces created by the up-and-down movements of the roller are absorbed by the table (anvil effect). Thus only a portion of these forces are actually transmitted to the axial thrust bearing of the gearbox.

- Grinding Plate and Clamping Ring

The grinding plate is the grinding track lining of mill. It is divided into segments clamped in position between the inner and the outer table collars via the clamping pieces of clamping ring. Each plate segment is associated with a specific clamping ring segment.

Migration of the grinding plate segments inside the table body will be prevented by means of drive pins.

For a detailed description of the grinding plate please refer to the "Grinding Parts cpl." section of Operating Instructions.

Dam Ring

The dam ring is bolted onto the outer collar of the grinding table body from above. The dam ring height determines the required thickness of the layer of grinding material on the table which is normally referred to as the grinding bed.

The unit of dam ring and lining ring is called "dam ring" in the following.

As a deep grinding bed leads to a higher energy consumption of the mill without increasing the grinding effect correspondingly, the dam ring should be kept as low as possible. At the same time, however, it must be considered that with a low dam ring less material will be held on the table so that the material thrown off by the centrifugal force will increase the load on the louvre ring. Dam ring height and specific grinding pressure will be optimized at the time of commissioning and initial startup.

When the mill is shipped the dam ring height is chosen so that it covers the special scope of application corresponding to our experience. The various qualities of the feed material such as grindability factor, feed size and feed moisture content demand in many cases a height adjustment of the dam ring when commissioning.

The dam ring consists of a solid base ring with machined collar for centering on the table collar. A package of flat steel rings is screw-fastened on the base ring bolted to the table. The overall height of dam ring above the grinding plate may be adjusted by removing or adding such flat steel rings.

It covers the large air gap between the outside diameter of table and the inner tire ring of louvre ring whereby a narrow gap will be formed in which a high flow resistance is prevailing. In this way, the hot gas will be forced to follow the path of lesser resistance, i.e. to stream through the louvre ring into the LM upper housing.

For faultless mill operation it is important that the rollers run as near as possible to the lower lining ring.

Scatter Ring (if existent)

The scatter ring has the task to distribute the grinding stock thrown over the dam ring and to feed it to the louvre ring.

- Louvre Ring

A stationary louvre ring encircles the table. It is a welded fabrication. Its function is to evenly guide the gas which flows out of the ring duct (see mill lower housing) into the grinding chamber. The angular position of the blades influences the circulation of the grinding stock within the chamber.

The gas velocity within the louvre ring is of special significance. It is a function of the primary gas volume and the cross-sectional free area of the louvre ring. Low flow velocities lead to a higher reject quantity through the louvre ring due to gravitational forces.

The segmental design of the louvre ring allows the segments to be exchanged positionally to increase their operational lifetime. Care must be taken to minimise the gap between the segments to avoid creating a gas bypass.

Hardfacing protects the outside of louvres from abrasive jet wear.

- Armoured Ring

The armour ring is the extension of the outer tire ring of louvre ring. It serves to impart the inclined gas stream emerging from the louvre ring in tangential direction an additional flow gradient in the direction of mill center. The superpositioning of both directions of flow will have a decisive influence

- on the internal particle flow within the grinding area,
- simultaneously on the grinding bed formation on the table and
- the availability of particles at the classifier mounted on the mill.

Each armour ring is made to suit the anticipated situation within the mill. The armoured ring is a welded fabrication whose components can also be positionally interchanged to minimise the effects of wear.

- Scraper

Foreign bodies and grinding stock particles which have fallen against the primary gas stream through the louvre ring or the grinding plate holes are collected on the floor of the ring duct which belongs to the LM-lower housing. Here, during operation or on starting of the mill, the pieces are caught by the scrapers and forced towards the reject funnel located in the primary gas inlet stud of the LM-lower housing.

2. Servicing

To maintain the mill in good condition and enhance the availability we suggest regular inspection and maintenance work for which the following information is given:

- Grinding Plates

The holes in the grinding plate have to be checked regularly on permeability. (s.a. Operating Instructions, "Grinding Parts cpl.")

- Table Body

During operation, the table body will reach a steady-state temperature in excess of 100 °C. Upon shutdown of the mill, reject flaps and mill door should not be opened fully, in order to prevent the table from getting in contact with large volumes of cold air which could result in considerable thermal stressing and possibly also in cracking of the table.

Attention should also be paid that no water will directly flow onto the grinding plate and the table, since stress cracking could also be caused thereby.

Another point to be observed is that the cold mill should be heated slowly as a function of standard temperature control. The setpoint of limit temperature downstream the LM classifier may not be changed unless approved by the LOESCHE staff.

- Armour Ring

The armour ring segments should be checked regularly for wear. Heavily worn and slightly worn segments should be exchanged with one another to avoid disturbances in the primary gas stream and, thereby, further damage.

- Dam Ring

A basic requirement for the smooth running of the mill is a constant dam ring height over the whole circumference of the table. Unevenness and dents caused by wear or foreign bodies should be attended to immediately if possible. In case of wear the complete plate ring set may be removed by unbolting and replaced by a new one. (see also "QA-Regulation: Dam Ring"; the number of which may be found in the parts list for Customer Documentation (DK)). The lower lining ring can be heavily worn and must be checked and replaced frequently.

When after approx. half the operational life of the grinding parts the effective height of the dam ring has increased due to wear of the grinding plate the dam ring can be reduced to its original height in order to achieve approximately the same specific power requirement (kWh/t) as during commissioning.

- **Louvre Ring**

The segmental design of the louvre ring allows the segments to be exchanged positionally to increase their operational lifetime.

- **Scraper**

The scrapers should be checked regularly for steadiness and stability of form. Inspection can be carried out through the gas duct with the reject flaps open.

When the scrapers are worn they must be renewed. Possible wear traces at the scraper mountings can be eliminated by armouring with hard electrodes.

Attention !

When performing welding work, connect negative pole of welding machine in direct vicinity of welding point and/or attach copper braid to the gearbox accordingly!

Grinding Parts cpl.

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1. General

The "Grinding Parts cpl." assembly is composed of the grinding plate (functional part of the "Table cpl." assembly) as well as the tires for the M- and the S-rollers.

2. Structure and Function

Grinding Plate

The grinding plate is the armoured part of the grinding track. It is a wear-resistant casting. The grinding plate is divided into segments, whose grounded contact surfaces are adequate support on the table. The segments will be clamped in position between the inside and the outside table collar.

The grinding plate is brittle as a result of its microstructure. In case of cracking of a segment due to mechanical or thermal stresses, this will not be a reason for replacement of the segment concerned as long as segment fragments will not migrate upward and the grinding track will not become perforated.

In general, cracked grinding plate segments will tightly wedge due to the penetration of grinding stock so that there is no need to replace them before expiry of their normal service life.

A regular exchange of the grinding plate is not foreseen. Due to anticipated concentrated wear in a limited zone it will be necessary to refurbish the groove formed by wear by welding. The application of weld material is accomplished by a mobile machine (see manuf. documentation) inside the mill itself. The grinding plate need not be removed. Based on our experience, rewelding should be done when the groove has reached 10 mm in depth.

M-Roller Tire

The roller tire is a wear-resistant casting.

Due to anticipated concentrated wear in a limited zone it will be necessary to refurbish the groove formed by wear by welding. The application of weld material is accomplished by a mobile machine (see manuf. documentation) inside the mill itself.

The way in which the tire is retained on the hub can be taken from the Operating Instructions "M-Roller cpl."

S-Roller Tire

The tire is a wear-resistant casting. The non-profiled surface has a working width corresponding to that of the M-roller.

The mode of fixation is described under "S-Roller cpl." in the Operating Instructions.

3. Maintenance

In order to ensure unobjectionable operation of the mill following routine inspections and maintenance procedures are recommended:

Grinding Plate

When no grinding stock is on the table, attention should be paid as a precaution that no water will run onto the table, in order to avoid thermal stresses and cracking.

When the grinding plates inside the mill are provided with lining, proceed as described in the Operating Instructions for Mobile Welder (see maker's documentation).

M-Roller Tire

A tire profile ruler will be supplied together with the "LM Auxiliaries cpl." assembly for checking the S-roller tire for wear. As regards structure and function please refer to "LM Auxiliaries: Roller-Tire Profile Ruler, cpl." in the Operating Instructions.

Replacement of M-roller tire should be effected according to "LM Auxiliaries: Roller - Lifter cpl." in the Operating Instructions.

When the roller tires inside the mill are provided with lining, proceed as described in the Operating Instructions for Mobile Welder (see maker's documentation).

S-Roller Tire

Replacement of S-roller tire should be effected according to "LM Auxiliaries: Roller-Lifter cpl." in the Operating Instructions.

Welding Work

Attention !

When performing welding work, connect the negative pole of welding machine in direct vicinity of welding point and/or attach the copper braid to the gearbox accordingly!

M-Roller cpl.

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1. Quantity and arrangement

The mill is equipped with two M-rollers, each of which is clamped in its own rocker arm. The rollers are standing opposite to each other. In the zero position the conical grinding surface stands parallel to the horizontal grinding path.

2. Construction

Each M-roller consists in essence of the fixed roller axle, rotating hub, rolling bearings, cover disc for roller axle, bearing sealings, roller tire as wearing part (see assembly "Grinding Parts cpl."), hub cover, sealing sleeve and lubrication system.

3. Bearings

Each M-roller is supported by a cylinder roller bearing which is arranged on the rocker arm side, and a dual-row tapered roller bearing on the cover side. These taper roller bearings (face-to-face arrangement) represent a functional unit with predetermined play. While the cylinder roller bearing absorbs radial forces only, the tapered roller bearing transmits both radial and axial forces. The axial forces are due to the M-roller being arranged inclined to the table. The two bearings are mutually supported through spacer sleeves and clamped by the screws of the roller hub cover and the sealing sleeve. Bearing clearance is defined by the seating on the roller axle and in the roller hub as well as within the tapered roller bearing, so no check measurements are required when mounting.

The size of the rolling bearings has been calculated to enable them to adequately withstand operational forces. However the actual service life depends to a significant extent on the way the mill is operated, the use of qualitatively required and clean lubricants, adequate lubrication (oil flow) and the maintaining of the prescribed mill operating temperature.

4. Lubrication

The rolling bearings are usually lubricated by an oil-recirculating system powered by the pumps of the hydraulic cabinet HSMS. A radial drillhole in the centre of the roller hub guarantees a constant minimal oil level within the roller.

(See operating instructions "Hydraulic cabinet HSMS")

5. Sealings

Each roller has a number of different sealings.

The annular gap gas-flushed seal prevents dust from the grinding space contaminating the roller bearings.

(See operating instructions "Sealing Air System cpl.")

The roller bearing chamber is at atmospheric pressure because it is connected with the exterior of the mill by a bore in the roller axle. This bore is equipped with an oil-wetted air filter. Because the pressure within the roller is higher than that of the mill interior no dust can get into the bearings.

The rolling bearings space is protected by mechanical sealings from the dust-loaded atmosphere in the mill as follows:

- An O-ring seals the hub cover relative to the hub.
- The sealing sleeve is screwed to the hub with the use of a sealing mass.
- The rotating parts of the roller are sealed relative to the non-rotating axle with a system of slip-ring sealings:

(see operating instructions "Slipring Sealing cpl.")

6. Adjusting of Rollers

The position of each roller relative to the table (distance to the dam ring) can be changed by means of a ring. This ring is inserted between the conical sleeve of the rocker arm and the relevant axle shoulder. The ring should not always be fitted but only when the roller contacts the inner diameter of the dam ring (see drawing).

7. Wear

The M-rollers are comminution tools and suffer wear.

7.1 Tire and Fixation Thereof

As with the grinding plate (see "Grinding Parts cpl.") the roller tire is a wear component. Since its metallurgical properties can be varied to suit requirements, this is explained in more detail in the Operating Instructions "Grinding parts cpl.".

7.1.1 M-Roller Tire Fixation

At the large diameter end, the v-groove sits on a corresponding collar of the hub. At the small diameter end the respective profiles are pressed together by a clamping ring.

The tensioning of that all is achieved by hammer-head bolts and nuts. They cramp the clamping ring with the roller tire and the roller hub. Thermal effects are compensated for by packets of spring washers under each nut.

7.1.2 Breaking and Fracturing Behaviour

Should a crack be induced during machine operation at one or more points, the clamping system will ensure that no parts of the tire fall out.

7.2 Bell Housing

The hub is protected against abrasive wear by a rotating bell housing, made of wear-resistant material.

A dismountable guard plate is connected with the bell housing for protecting the rocker arm sealing against abrasive wear.

8. Servicing

Regular and thorough maintenance of the rollers increases their service life and at the same time the availability of the mill. Particular attention should be paid to the following points:

- Ensure that the mill runs smoothly (make sure that the slides and flaps in the gas lines close tightly; check the height of the dam ring ...).
- Keep the height of the dam ring relative to the grinding path exactly uniform around the complete perimeter; set the roller force correctly in accordance with the springs diagram.

- The efficiency of the sealing air can be checked whilst the mill is in operation:
 - Mills without a sealing air fan by laying the palm of the hand over the air inlet bore to the sealing air chamber on the M-rocker arm. One feels a distinct suction effect.
 - On mills with a sealing air fan by removing the plug in the end of the oil leakage hose; pressurized seal air will then escape here.

If this is not the case, either the annular gap between the sealing air ring and counter ring on the M-rocker arm has become too great as the result of wear (see drawing details) or the sealing air chamber in the region of the sealing sleeve and M-rocker arm is contaminated with dirt.

In the first case the 2-part rings and the counter rings, which together form the sealing air gap, should be replaced; in the second case the above-mentioned chamber should be cleaned as soon as possible after first removing the complete M-roller since otherwise there is acute danger of the rolling bearings being damaged.

- When dismantling the complete M-roller, the latter should be swung out of the mill together with the M-rocker arm. The clamping sleeves can then be released in accordance with the "LM-Auxiliaries: M-Rocker Arm" operating instructions.

When reassembling, take care that the slits are positioned in the conical clamping sleeves in accordance with the drawing! The slit in the taper sleeve must be caulked with lead so that it is gas-tight in accordance with design drawing "Rocker Arm (M) cpl." so that no sealing gas can get out or dust get in through it.

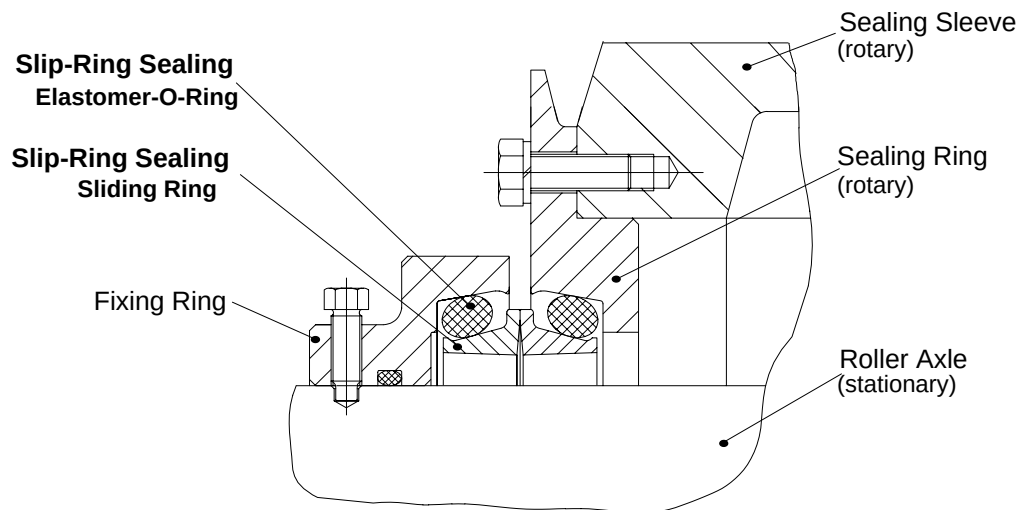
- Slip-ring sealing (see operating instructions "Slip-ring sealing cpl.").
- For occasional checking of the tapered roller bearing, the inspection cover in the hub cover can be removed. When replacing the latter, take care that it seals adequately and that the bolts are retained by welding wire straps.
- If the hub cover should be removed to permit the rolling bearings to be checked, the M-roller must be brought into a vertical position with the aid of the swinging-out device. Prior to doing this, seal the holes in the roller axle in accordance with the details in the drawing to prevent oil getting out through it. The O-ring must always be renewed before refitting the hub cover. Only moulded O-rings as specified in the parts list may be used! The cover screws must be tightened up and locked in accordance with the details in the drawing.
- All parts must be checked regularly for impingement wear at regular intervals and replaced if necessary. Repairing of parts by welding should only be carried out - in so far as this is possible - following consultation with the manufacturer since otherwise breakage of the components can result.

Roller: Slip-Ring Sealing cpl.

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1. Structure



principle sketch

2. Function

The slip-ring sealing has the task of keeping the lubricant in the roller bearing chamber of the grinding roller. Therefore the rotating parts of the roller are sealed against the static axle by a slip-ring system. When they are new, the two rings sliding on one another are lapped. Under elastic load they do not let any lubricant through.

It is not the task of the slip-ring sealing to keep dust away from the roller bearings. This is rather the task of the sealing air system (see operating instructions "Sealing Air System"). Extreme fouling in the roller bearing chamber consisting of grinding material always shows inadequate functioning of the sealing air system.

3. Servicing

- As the slip-ring sealing is subject to wear it has to be renewed regularly (at each major inspection at the latest) thus avoiding damage to the roller bearing due to leakage. The condition of the slip-ring sealing can best be indirectly checked. When swinging out the rollers the oil in the roller bearing chamber runs through the center bore of the axle downwards. It can be led to a collecting pan (to be provided by the customer). When the drained oil shows strong discoloration because of microscopical suspended particles there is a possibility of the slip-rings being worn. In this case the roller must be dismantled and the condition of the slip-rings must be checked.
- N. B. temperature limit

If the contractually agreed temperature behind the mill is exceeded it can lead to hardening of the Elastomer O-rings and subsequent premature failure of the seal.
- The slip-rings must be replaced in pairs.

M-Rocker Arm cpl.

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1. Quantity and Arrangement

The mill is equipped with two M-rocker arms (M="Master"), each of which is mounted in a bearing block of the LM-lower housing. The rocker arms stand transposed to one another by 180°.

2. Structure

Each M-rocker arm consists primarily of the central arm, fork, axle, and radial as well as axial bearings in the bearing block of the LM-lower housing.

The central arm is shrunk onto the axle. The exact axial position of the arm on the axle is determined by means of a shoulder. In the upper portion of the arm at right angles to the axle is the bore for the roller axle. The roller axle is fixed by means of conical sleeves.

The fork grips the rocker arm from both sides. It is also fixed on the axle. The connection to the axle is achieved with conical flanged sleeves which have only the purpose to hold the fork and axle concentrically to one another. This clamping connection does not transmit torque. The torque is transmitted via a screwed and pinned connection (see drawing "Rocker Arm (M) cpl."). Each cheek of the fork is pulled up against the central arm by means of bolts. The torque is transmitted through friction between the fork and central arm and also through the positive connection provided by an expanding sleeve and pin (taper pin connection) in each cheek of the fork.

The outside of the slotted clamping sleeve is cylindrically machined and the bore is tapered. The clamping sleeve is inserted into the reamed bore of the fork and central arm. The tapered pin is forced into the clamping sleeve by means of a set of disc springs, the cover and screws, thus expanding the sleeve and forcing it against the reamed bore of the arm and fork (see operating instructions "LM-Auxiliaries: Rocker Arm - Taper Pin Connection").

The lower portion of the fork is a box section which terminates in two lugs with a right angled bore. This bore accepts the pin for mounting the rod head which connects the double-acting hydraulic cylinder of the hydropneumatic spring system to the fork.

The fork possesses a bolted on plate at the position where the transverse bore passes through both cheeks. This is the point at which the complete rocker arm presses against a flexible buffer (see assembly group "Spring Assembly cpl.". With the help of this device the travel of the rocker arm and roller system is mechanically limited to prevent contact with the table.

Note: On the front face of the rocker arm where the roller axle is clamped with double-conical sleeves, there is a sealing box mounted. This box, which is armoured, forms a part of the seal where the M-rocker arm penetrates the mill body (see assembly "Rocker Arm Sealing (M) cpl.").

3. Bearings

The M-rocker arm axle is supported by two bearings on the bearing pedestal, which is part of the M-mill stand of the LM-lower housing. The radial forces are absorbed by two spherical roller bearings, which permit the axle of the M-rocker arm to swivel. The thrust transmitted from the table via the M-roller to the M-rocker arm is absorbed by thrust rings. Two of these thrust rings are mounted on the axle and one, which lies between them, is bolted to the bearing housing.

During the grinding operation the M-rocker arm and its axle are pushed in the direction in which the table rotates. Should this thrust on the M-rocker arm be momentarily relieved, and the rocker arm should recoil, thrust will be felt in the opposite direction. The axial forces will then be alternatively directed to the housing side or the axle side thrust rings. The thrust rings are prevented from rotating relative to the axle and the bearing housing of the LM-lower housing by roll pins.

Peak loads which are led into the bolts of the bearing housing are dissipated by disc springs and friction.

4. Function

The M-rocker arm has the task of transmitting the force produced by the hydraulic double acting cylinder of the assembly "Spring Assembly cpl." into the roller and the grinding bed. The force is thereby led from a jointed pin for the spring rod in the lower part of the fork via the split sleeve/pin joints into the hubs, the central rocker arm and the rollers. The torque arms for transmitting these forces (distance between swivel pin in the fork and the rocker arm bearing / rocker arm bearing to the centre point of the roller) are so dimensioned that the definite but low bearing reaction forces are created.

The separation of the central rocker arm and the fork allows to keep the fork and hydraulic cylinder of the springing device in position during the swinging-out process of the rocker arm/roller unit. Dismounting of these components is not necessary.

The internal sealing system between the roller and rocker arm is protected against the ingress of dust by an air seal. The air is fed to a coaxial gap between the roller and the rocker arm via a bore in the middle of the rocker arm. At the air inlet the protective cap (used for transport) is exchanged for a seal-air inlet pipe (see operating instructions "Sealing air system").

Note: The sealing between the rocker arm and the mill body is achieved via the sealing box with its vertical sides and rounded upper and lower surfaces. The rounded surfaces rotate concentrically about the M-rocker arm axle (see "Rocker Arm Sealing (M) cpl.).

5. Lubrication

Grease lubrication has been provided for all bearing assemblies. During shipment and erection all lubrication connections are closed with grease nipples. Upon completion of erection, all grease nipples are removed and the lubrication points are connected to a centralized lubrication system. For further details see operating instructions "Lubrication Line cpl.".

Only high performance lubricants are to be used. Specifications are shown in our servicing check-list.

6. Servicing

The M-rocker arms as well as the grinding rollers are the precision parts of the mill. Their value can best be retained through proper servicing.

The following should be observed:

- Arm/fork connection:

see operating instructions "LM-Auxiliaries: Rocker Arm - Taper Pin Connection"
- The clamping connection between the fork and the axle must be tight. Even a small amount of clearance will cause deformation of the bore in the fork and could lead to eventual breakage of the fork.
- The bolts of the axial thrust rings must be checked for tightness. The cap must be removed for this purpose.
- All bearings are to be lubricated regularly. An interruption of the flow of lubricant will lead to seizure of the bearings (see operating instructions "Lubrication Line cpl.>").
- To transport the M-rocker arm lugs are attached. They are a part of the supply and are shown in the group assembly drawing. The lugs are to be removed before commissioning.
- The leakage-oil hose (transparent) - if present - leads from the sealing air chamber down underneath the rocker arm axle:
 - The hose should be inspected to see if there is any oil present inside it. Should leakage oil be present here, then the slipping seal of the roller should be checked.
 - Before swinging the roller out of the grinding chamber the clamping point of the hose on the pedestal must be loosened (see drawing "Rocker Arm cpl.>").

When reinstalling, avoid loops in the hose so that leakage oil can flow into the bottle mounted below the hose clamp.
- The striker plate on the lower end of the fork must be exchanged if it becomes deformed through frequent contact with the buffer.

M-Rocker Arm: Sealing cpl.

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1. General notes

The group "Rocker Arm Sealing (M) cpl." is comprised of moving parts which are mounted on the rocker arm and stationary parts which are bolted to the upper housing.

2. Structure

2.1 Moving Parts

The parts which are mounted on the M-rocker arm are composed of

- the sealing cap with armoured side walls (see section),
- the two part armoured ring,
- an upper armoured plate on the sealing cap and a lower sealing segment which is bolted to the sealing cap,
- and
- an armour ring which is bolted to the front face of the sealing cap.

2.2 Static Parts

The stationary parts are bolted to the LM-upper housing cover and the LM-upper housing itself. These parts are as follows

- an upper sealing bar and a lower sealing bar which have curved surfaces to suit the moving sealing segments,
- two right hand and two left hand "side walls" (angle strips),
- various rubber and plate strips which are bolted to the faces of the sealing bars.

3. Function

By means of elongated holes the rubber and plate strips may be adjusted such that they almost touch the moving parts (sealing box). This is also true for the sealing segments and the angle strips.

The adjustable strips create long and narrow labyrinth gaps between the moving and stationary parts. The gaps prevent dust penetrating to the externally mounted sealing strips. The sealing strips, which are made of fabric reinforced rubber, can be produced from old rubber conveyer belts.

For sealing the right angled corners between the sealing cap and the cover of the mill body, special corner seals are bolted on to minimize the entry of secondary air into the mill.

The sealing system is operative

- when the roller is lifted before the mill starts,
- when the roller is in its operating position and
- when the roller is in a worn condition.

4. Servicing

- Maintain the smallest possible gap between the stationary and the moving parts.
- The rubber strips must always be sandwiched between two steel strips. The rubber should project only 2 mm (0.1 in) in front of the metal strips. Larger projections mean that the rubber will be bent backwards and forwards and will break due to fatigue.
- The two part armour ring should be replaced when it is worn to avoid the sealing air system from becoming worn itself.
- The armour plates of the sealing box should be regularly checked for wear. They should be replaced when worn before the basic parts of the sealing box also suffer wear. The armour plates are made of wear resistant material.

Rocker Arm Monitoring cpl. with Grinding Bed Control

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1. General

The rocking angle of the rocker arm determines the roller position with the two proximity switches limiting the rocking range of the rocker arm. In this context the upper limit serves to switch off the hydraulic system after the hydraulic lifting, the lower limit serves to prevent metallic contact between the grinding elements through activation of the lifting process. In addition, the current rocking angle of the rocker arm is continually monitored by an inductive displacement transducer. It is a measure of the grinding bed thickness and is also indicated on a scale attached to the rocker arm.

2. Function of the proximity switches

The limiting positions of the rocker arm are monitored by means of electronic proximity switches which are mounted on the sides next to the fork below a protective cover on the bearing housing of the rocker arm. The two limiting positions "Roller lifted off" and "Grinding bed too thin" are indicated by different proximity switches, i.e. each limiting position is monitored by its own proximity switch.

Long holes allow the proximity switches to be adjusted precisely for each position. The proximity switches are activated when no attenuation is experienced by opposing metallic surfaces. They are formed by metal plates attached to the sides of the rocker arm.

The inner proximity switch (with respect to the mill centre) is activated when the roller is lifted. The outer proximity switch is, depending on the wear of the grinding elements, adjusted in such a way, that it is activated shortly before the grinding elements make contact.

3. Switching off in the limiting position for "Roller lifted off"

When the proximity switches for all the rollers connected to hydraulic system 1 (HSLM), signal "Rollers lifted off", the hydraulic pump of the HSLM is to be switched off.

4. "Grinding bed too thin" monitoring

If one or several rollers repeatedly or continually signal "Grinding bed too thin" the function "Lift rollers" for the hydraulic system (HSLM) must be activated in order to prevent mechanical damage.

CAUTION The adjustable plates for the outer proximity switches (with respect to the mill centre) must be readjusted to compensate for the increasing wear of the grinding elements, as a different rocker arm angle is required with worn grinding elements than with new ones. Without readjustment the system would activate the function described above even though the grinding bed thickness had not yet fallen below the minimum thickness.

5. Monitoring the grinding bed thickness

The grinding bed thickness can be determined directly from the angle of the rocker arm. This is measured by means of an inductive displacement transducer with the aid of a wedge-shaped position transmitter. The displacement transducer is mounted onto a bracket attached to the bearing housing, while the position transmitter is screwed to the lower end of the holding plate connected to the rocker arm.

The active area of the displacement transducer loses energy when a metallic body is placed close to it. This energy loss is used to measure the distance between the position transmitter and the sensor. The wedge-shape of the position transmitter, covering the complete lift of the roller, causes the distance to change continuously with the roller movement, so that each roller position is correlated to a certain distance of the sensor from the position transmitter.

The energy loss experienced by the sensor is transmitted to the process control system as an analogue signal. If possible, these signals should be calibrated with values corresponding to the grinding bed thickness (for example in mm). As the grinding elements wear, this calibration must be renewed.

In addition, at the mill, the grinding bed thickness is indicated visually, directly onto a scale. In order to achieve this, a bracket with an adjustable pointer is screwed to every LM upper element cover. The pointer points to a "scale" on the rocker arm representing the grinding bed thickness at a magnification of 1:1. Here again, the zero position of the pointer has to be readjusted as the grinding elements become worn.

S-Roller cpl.

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1. Quantity and Arrangement

The mill is equipped with two S-rollers, each of which is clamped in its own rocker arm. The rollers are standing opposite to each other. In the zero position the conical grinding surface stands parallel to the horizontal grinding path.

In order to achieve an optimal rolling effect, the inclined roller axle and the vertical table axle will coincide in the working plane.

2. Construction

Each S-roller consists in essence of the fixed roller axle, rotating hub, rolling bearings, cover disc, bearing sealings, roller tire (see assembly "Grinding Parts cpl."), hub cover and sealing sleeve.

3. Bearings

Each S-roller is supported by a cylinder roller bearing which is arranged on the rocker arm side, and a spherical roller bearing on the cover side. While the cylinder roller bearing absorbs radial forces only, the spherical roller bearing transmits both radial and axial forces. The axial forces are due to the S-roller being arranged inclined to the table.

4. Lubrication

The roller bearings are lubricated in an oil-bath. Please see the lubricants table for the quantity and quality of oil required.

5. Sealings

Each roller has a number of different sealings.

The annular gap gas-flushed seal prevents dust from the grinding space contaminating the roller bearings.

(See operating instructions "Sealing Air System cpl.")

The roller bearing chamber is at atmospheric pressure because it is connected with the exterior of the mill by a bore in the roller axle. This bore is equipped with an oil-wetted air filter. Because the pressure within the roller is higher than that of the mill interior no dust can get into the bearings.

The rolling bearings space is protected by mechanical sealings from the dust-loaded atmosphere in the mill as follows:

- An O-ring seals the hub cover relative to the hub.
- The sealing sleeve is screwed to the hub with the use of a sealing paste.
- The rotating parts of the roller are sealed relative to the stationary axle with a system of slip-ring sealings:

(see operating instructions "Slipring Sealing cpl.")

6. Adjusting of Rollers

The S-roller is no grinding roller but a preparation roller. It serves the purpose of pre-compacting the aerated and non-homogeneous grinding bed composed of grinding stock charged and recirculated grinding stock, i.e. of preparing the bed for the downstream grinding roller (M-roller).

It is for this reason that the working plane of S-roller is substantially higher than that of the M-roller. The position of S-roller above the table surface may be selected in two different ways:

- by adjustment of the S-buffer spindle in that the S-roller rocker arm complete with S-roller is mechanically adjusted to a desired height;
- by means of the S-hydraulics by way of positioning via a hydraulic cylinder.

7. Wear

The S-rollers serve for preparation of the grinding bed. They work with low specific pressures and theoretically do not exert any sliding movement on the grinding bed. Thus, roller wear is minimum.

7.1 Tire and Fixation Thereof

The S-roller tire belongs to the "Grinding Parts cpl." assembly and is described under the same designation in the Operating Instructions.

7.1.1 S-Roller Tire Fixation

The S-roller tire comprises two cylindrical seats of different size on the S-roller hub. The tire will be clamped in position by means of a clamping ring and various bolts to abut against a plane shoulder of the hub.

7.2 Bell Housing

The hub is protected against abrasive wear by a rotating bell housing, made of wear-resistant material.

8. Servicing

- For disassembly of the S-roller, the latter will be swung to the outside together with the LM upper housing cover (see "Swinging Device cpl." assembly).
- When refitting an S-roller in the S-rocker arm, make sure to pay attention to the position of slots in the conical clamping sleeves as shown in the drawing! According to the design drawing ("S-Rocker Arm cpl.") the slot in taper sleeve must be made gastight by caulking with lead, in order to prevent sealing gas from escaping and/or dust from penetrating.
- Slip-ring sealing (see operating instructions "Slip-ring sealing cpl.").
- To avoid damage to the roller bearings it is necessary to check regularly the effectiveness of the air seals.

(see operating instructions "Sealing Air System")

- Check the oil level with the aid of the oil dipstick supplied. The latter has two marks. The oil should be topped up to the lower mark when the grinding parts are new and to the upper one when they have become worn. The oil should in any case be topped up to the upper mark not later than after 3000 operating hours.
- For occasional checking of the spherical roller bearing, the inspection cover in the hub cover can be removed. When replacing the latter, take care that it seals adequately and that the bolts are retained by welding wire straps.

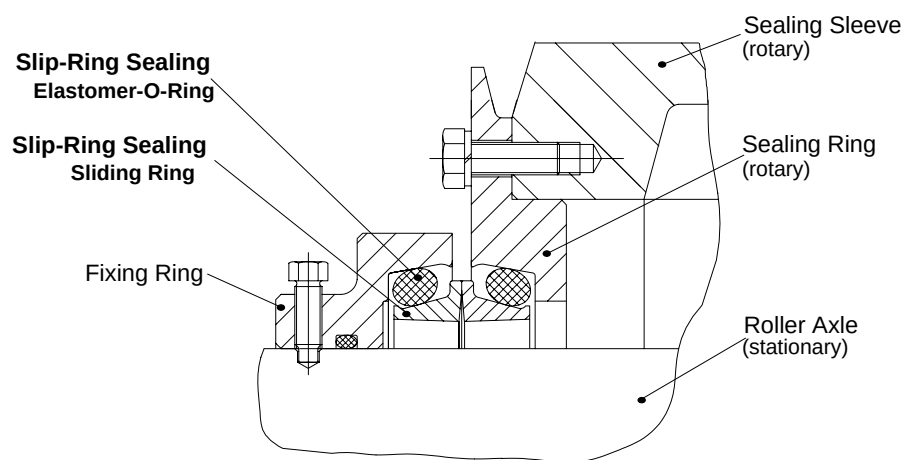
- If the hub cover should be removed to permit the rolling bearings to be checked, the S-roller must be brought into a vertical position with the aid of the swinging-out device. Prior to doing this, seal the holes in the roller axle in accordance with the details in the drawing to prevent oil getting out through it. The O-ring must always be renewed before refitting the hub cover. Only moulded O-rings as specified in the parts list may be used! The cover screws must be tightened up and locked in accordance with the details in the drawing.
- All parts must be checked regularly for impingement wear at regular intervals and replaced if necessary. Repairing of parts by welding should only be carried out - in so far as this is possible - following consultation with the manufacturer since otherwise breakage of the components can result.

Roller: Slip-Ring Sealing cpl.

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1. Structure



principle sketch

2. Function

The slip-ring sealing has the task of keeping the lubricant in the roller bearing chamber of the grinding roller. Therefore the rotating parts of the roller are sealed against the static axle by a slip-ring system. When they are new, the two rings sliding on one another are lapped. Under elastic load they do not let any lubricant through.

It is not the task of the slip- ring sealing to keep dust away from the roller bearings. This is rather the task of the sealing air system (see operating instructions "Sealing Air System"). Extreme fouling in the roller bearing chamber consisting of grinding material always shows inadequate functioning of the sealing air system.

3. Servicing

As the slip- ring sealing is subject to wear it has to be renewed regularly (at each larger inspection at the latest) thus avoiding damage at the roller bearing due to leakage. The condition of the slip- ring sealing can best be indirectly checked. When swinging out the rollers the oil in the roller bearing chamber runs through the center bore of the axle downwards. By means of an oil-draining device to the roller (see operating instructions "LM-Auxiliaries: Roller - Oil Drain Device cpl.") it can be lead to a collecting pan (to be ordered by the customer). When the drained oil shows strong decoloration because of microscopical suspended particles there is a possibility of the slip-rings being worn. In this case the roller must be dismantled and the condition of the slip-rings must be checked.

The slip-rings must be replaced in pairs.

S-Rocker Arm cpl.

Table of Contents:

	Sheet
1. Quantity and Arrangement	1
2. Structure	1
3. Bearings	1
4. Function	2
5. Lubrication	2
6. Servicing	2

1. Quantity and Arrangement

The mill is equipped with two S-rocker arms (S = "Slave") each of which is supported in pedestal bearings of the LM upper housing. The rocker arms are offset from each other under an angle of 180 degrees.

2. Structure

The complete S-rocker arm is substantially composed of the arm, the axle, two pedestal bearings, and armouring parts.

The arm is a T-shaped casting consisting of two tubular cross elements. Conical sleeve sets for accommodation and fixation of the S-roller axle are arranged at both ends of the longitudinal element. A conical sleeve is provided as a taper sleeve at the end located at the top during operation to permit disconnection in case of dismantling.

On both sides, the transverse tube terminates in machined journals over which said pedestal bearings are positioned.

Critical zones of the S-rocker arm subjected to abrasive wear are protected by wear resistant steel armouring. For details please see assembly drawing.

3. Bearings

The S-rocker arm is directly retained in two bearings (without additional axle). These bearings are each composed of a bearing housing bolted to abut against the S-rocker arm cover on the LM upper housing and a self-aligning roller bearing with grease packing. The bearing downstream the rocker arm/roller axle as seen in the direction of table rotation is designed as a fixed bearing (supplementary rings).

The radial forces will be taken up by both self-aligning roller bearings. The thrust transmitted from the table to the S-rocker arm via the S-roller will be taken up by the fixed bearing.

4. Function

The S-rocker arm serves

- for pivotably fastening the S-roller,
- for flexibly supporting the S-roller on the S-buffer and for holding it at the defined minimum clearance above the grinding track,
- for transmitting hydraulic ancillary forces resulting from the action of hydraulic cylinder engaging the S-rocker arm on its upper side into the S-roller.
- for sealing the S-rocker arm/roller unit at the housing feed-through via a flanged sealing shield (see "S-Rocker Arm Sealing cpl.").

Sealing air is introduced into the ring gap between roller axle and S-rocker arm via the hollow journals serving for internal sealing of rocker arm/roller to prevent the penetration of dust into the bearings of roller. To this effect, the transport protective cap must be replaced by the sealing air connecting sleeve in the planned connection point (see Operating Instructions, "Sealing Air System").

5. Lubrication

Rolling bearings are grease packed.

As regards appropriate relubrication intervals please refer to maintenance schedule and/or lubricant chart.

6. Servicing

The S-rocker arm is a precision part of the mill for which the appropriate maintenance procedures should be followed carefully.

In particular, attention should be paid to the points listed below.

- Bearing housing bolts need to be checked for proper seating despite being secured in place.
- Make sure to observe the appropriate relubrication intervals and grease qualities for rolling bearings.
- Change stop plate at the rocker arm in case of severe deformations after extended periods of operation.
- Inspect armouring elements for wear at regular intervals and replace them in due time to prevent damaging of the S-rocker arm.

S-Rocker Arm: Sealing cpl.

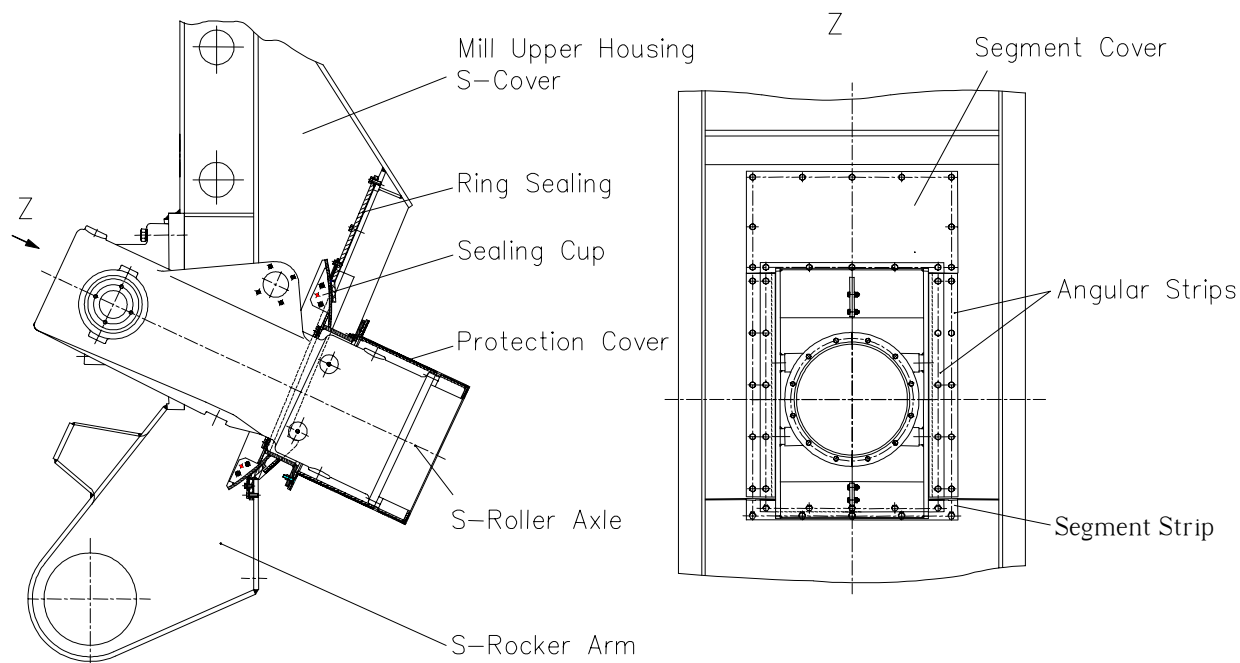
Table of Contents:

	Sheet
1. General Remarks	1
2. Structure	1
2.1 Moving Parts	2
2.2 Stationary Parts	2
3. Function	2
4. Maintenance	2

1. General Remarks

The "S-Rocker Arm: Sealing cpl." assembly is composed of moving parts fitted to the S-rocker arm and of stationary parts bolted to the LM upper housing.

2. Structure



principle sketch

2.1 Moving Parts

The parts fitted to the rocker arm comprise

- the split type sealing case made of steel sheet and the wear resistant protection cover
- the associated rubber strip sealing ring for the rocker arm collar.

2.2 Stationary Parts

The stationary parts are bolted to the S-cover for the LM upper housing. These parts are

- the segment cover placed on top of the sealing case,
- the segment strip extending over the sealing case from below,
- two (2) angular strips arranged on the right and on the left in parallel with the sealing case surfaces,
and
- seals (rubber sealing strips) abutting against the sealing surfaces of sealing case together with steel strips.

3. Function

The rubber and sheet metal strips can be readjusted up to the moving parts (sealing case) via elongated holes. The same applies to the sealing segments and the angular strips as such.

Narrow labyrinth gaps are formed between the moving and the stationary parts by way of the adjustable strips. They prevent the penetration of dust into the exterior sealing strips. The sealing strips consisting of fabric-reinforced rubber may be obtained from defective rubber belts.

4. Maintenance

- Pay attention to a narrow gap between moving and stationary parts.
- Make sure that rubber strips will always be supported by metal strips on both sides. The rubber strip should protrude by not more than appr. 2 mm from between the metal strips. Any protrusion in excess thereof means flexing work with the risk of rupturing due to fatigue.
- In case of heavy wear the sealing case should be replaced as a complete unit. Since of split design, the case may be removed by unscrewing with the S-roller swung out (lateral fastening screws on S-rocker arm and bolts in the center of parting flange).

S-Rocker Arm: Monitoring cpl.

Table of Contents:

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1. General	1
2. Monitoring of "Rollers lifted"	1
3. Monitoring of "Rollers too low"	1

1. General

The travel limits of the S-rocker arm are monitored by switches attached to the mill upper housing cover at the side of the rocker arm itself. The switches are placed at different locations and have different functions. They emit a signal when their activating surface is covered by (but not in contact with) a special adjustable rail which is attached to the rocker arm.

The position at which the switches are activated can be adjusted by moving the switches within their elongated mounting holes.

The signals mean either "Roller lifted" or "Rollers too low".

2. Monitoring of "Rollers lifted"

The innermost switch has the function of monitoring the lifted condition of the roller during the start-up phase of the mill. When all the rollers have been lifted, the switches give a signal to the PCL so that after a certain time period the hydraulic pump can be switched off.

3. Monitoring of "Rollers too low"

The outermost switch monitors the roller condition "Rollers too low" during mill operation. The adjustable rail is positioned so that no signal will be generated as long as the grinding bed has an adequate thickness. Should the grinding bed become too thin, the signal causes the control system to stop the mill so that the stop buffers are not additionally loaded with the hydraulically created force.

ATTENTION The slotted rails must be adjusted during mill operation to cater for wear of the grinding parts because the attitude of the rocker arm is different with worn grinding parts compared to new ones.

S-Roller: Stroke Monitoring cpl.

Table of Contents:

	Sheet
1. General	1
2. Structure	1
3. Function	1
4. Commissioning/Startup and Operation	1
5. Maintenance	1

1. General

The stroke monitoring system is operated by a displacement sensor which can give the roller's position in the mill.

2. Structure

The displacement sensor is mounted on a bracket which is fixed to the mill upper housing cover. The activating plate is either attached directly to the rocker arm or attached to it via a bracket.

3. Function

The working surface of the displacement sensor emits an inductive field which reacts with metallic components which enter this. The disturbance of the field is a measure of the gap between the activating plate and the sensor. The activating plate is wedge-shaped and thus the gap varies continuously over the entire travel of the roller.

Therefore to every roller position is allocated a distance.

4. Commissioning/Startup and Operation

The field strength of the displacement sensor is transmitted to the control system as an analogous signal these signals are not integrated further into the control system.

The analogous values should be calibrated as a function of distance (e.g. in mm).

5. Maintenance

The relative position of the sensor and activating plate must be readjusted to compensate for wear of the grinding elements.

Lubrication Line cpl.

Table of Contents:

	Sheet
1. General	1
2. Construction and Function	1
2.1 Oil Piping	1
2.2 Grease feed lines	2
2.3 Assembly	2
3. Servicing	2

1. General

This subassembly includes the oil piping for the circulating oil lubrication of the M-rollers and the grease feeds to the M-rocker arm.

2. Construction and Function

2.1 Oil Piping

Oil piping assembly as per subassembly drawing "Grease Line cpl." between connecting pipe of the hydraulic cabinet HSMS and M-rollers. Marking of the feed connection, see subassembly drawing or lubrication diagram.

Subsequent cross-section reductions in the feed pipes can lead, particularly in the return flow to the tank, to restriction of the function of the lubricating system.

At the roller axle the connection is made with hoses which allow free movement both during operation of the mill when the M-rocker arm is moved and during the traversing process.

2.2 Grease feed lines

The grease feed system as per assembly drawing "Grease Line cpl." runs between one or more motorized grease pumps (for number see parts list) and the appropriate lubrication points. These include the radial and axial bearings of the M-rocker arm.

Piston distributors with positive control allow an exact distribution of grease to the lubrication points. The piston distributors can only be guaranteed to function correctly when all outlets to the users are attached.

2.3 Assembly

All feed pipes are to be laid in protected positions under the platform or on the carriers of the platform outside the access area and mounted at appropriate points so as to be free of vibration.

The piston distributors must not be mounted on an uneven surface, or the free movement of the pistons will be impaired.

NOTE: All greasing points supplied from the central greasing unit are closed with flat lubrication heads during transport. These grease nipples can be used to supply each lubrication point with grease from a hand pump before the feed pipes are laid. When the greasing unit is connected, the greasing nipples must be replaced with screwed connections as shown in the drawing or parts list.

3. Servicing

Mechanical damage to the pipes must be remedied at once. Hoses are to be checked regularly to ensure that they are in good condition and replaced if necessary.

The grease container of the lubricating pump is to be refilled with the grease listed in the lubrication table at regular intervals. The lid of the grease container is to be shut tight, to prevent the entry of dust.

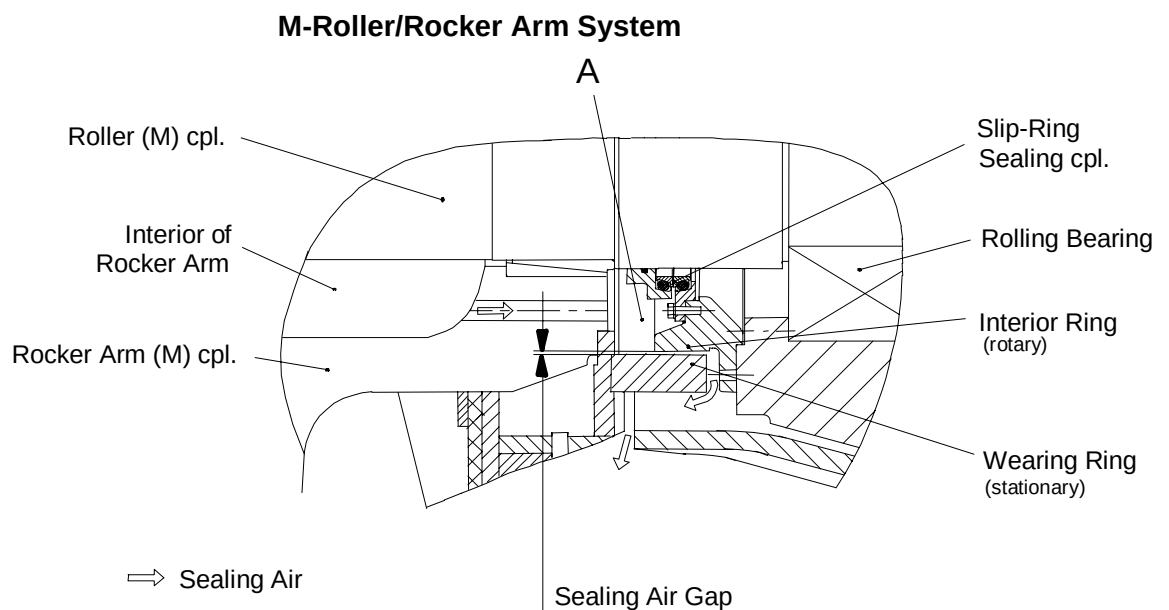
Sealing Air System

Table of Contents:

	Sheet
1. Structure and Function	1
2. Servicing	2

1. Structure and Function

To avoid damage occasioned by the ingress of dust from the grinding chamber into the roller bearings and the slip-ring sealing of the M- and S-rollers the small annular gap between the rotating interior ring of the rollers and the wearing ring of the stationary M- and S-rocker arm, which surrounds the neck of the roller, is pressurized with sealing air.



principle sketch

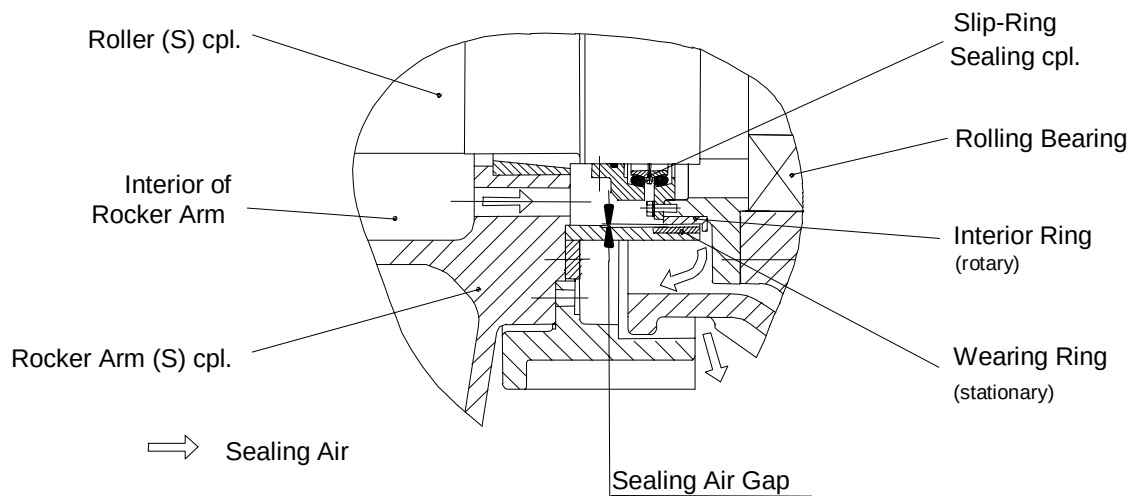
The material feed takes place through the sealing air piping surrounding the LM upper housing, to

- a borehole located in the upper part of the M rocker arms
- or
- a borehole located on the side of the S rocker arms

into the internal space of the rocker arms from where the air flows into the mill interior across the sealing air gap.

Due to the vacuum in the mill interior, the atmospheric air is drawn through the sealing air gap. In the case of an insufficient pressure level, additional use is made of a separate sealing air fan which is connected to the inlets of the rocker arms by means of a special piping system.

S-Roller/Rocker Arm System



principle sketch

The effectiveness of the sealing air gap is influenced by the velocity with which the air is sucked respectively pressed into the mill interior. If any wear becomes visible in the sealing air gap, the wearing rings fitted both on the rocker arm and on the roller are to be replaced.

2. Servicing

The effectivity of the air seal can be checked during a mill inspection by removing the roller and measuring the diameter of the wearing rings on the roller and rocker arm. The difference between them may not exceed the gap quoted in the drawing "Roller cpl.", otherwise the rings should be renewed.

Vibration Monitoring cpl.

Table of Contents:

	Sheet
1. General	1
2. Construction and Function	1

1. General

The evaluation of mill vibration is based upon VDI Guideline 2056. LOESCHE Roller Mills are placed into category "D" of the VDI Guideline. This group covers rigidly mounted machines without unbalanced mass effects. Because of the unbalanced mass effects there are no grading rules. For this reason, the mill may be reclassified into group "G" which covers rigidly mounted machines (on heavy foundations) with only rotating masses.

2. Construction and Function

To monitor the mill's running performance, a vibration pickup is mounted above the mill door on the body, according to the drawing "Vibration Monitoring cpl.". The vibration pickup measures in a horizontal direction the movement of the mill body (mm/s). The values measured are not foundation vibrations which are those which the VDI Guideline 2056 attempts to classify. Indeed, at the height of the mill door a very much larger vibratory speed will be registered compared to the magnitude in the foundation. The measured mechanical values are converted to electrical signals and transmitted via a cable to an indicator in the electrical cabinet where they are displayed. As soon as the indicator exceeds a predetermined value, the normal shutdown process of the mill is instigated.

Grinding bed sprinkler cpl

Table of Contents:

	Sheet
1. General	1
2. Construction and function	1
3. Pump capacity and flowrate	2
4. Maintenance	2

1. General

The grinding bed sprinkler is a feature with which the properties of the grinding bed can be modified. It has been shown in many clinker and slag grinding plants that sprinkling water onto the grinding bed can stabilise it and lead to a quieter running mill.

In front of each roller water is introduced from above onto the grinding bed. The quantity of water is so small that it exerts no appreciable influence on the thermal household of the mill and thus has no disadvantageous effect upon the process. The effect of the water is greater the nearer it is to the roller. However it must be considered that the water, should it come into contact with the roller tire, can destroy it. Thus the water is introduced at a safe distance in order to protect the tire.

2. Construction and function

The grinding bed sprinkler assembly consists of the pumping station and the pipework from the base of the mill pedestals to the spray tubes in the mill body, including the fixing elements.

The water piping is connected to the spray tube by a hose with a simple coupling. Thus the spray tube can be easily disconnected when its inspection or exchange is necessary. At the other end of the hose an orifice is installed into the piping to raise the pressure and maintain an even distribution of water to both of the spray tubes.

In the event that the mill should run for a longer period without the sprinkler system, the complete spray pipe and its support can be removed and the aperture in the mill body closed with the cover supplied.

3. Pump capacity and flowrate

The orifice is designed so that when the pump works at full power the maximum flow rate (see assembly drawing) will be achieved. When a frequency controlled pump is used, a minimum flowrate must also be observed as otherwise the pressure at the outlet may become too low and negatively affect the function. By changing the orifice (see suppliers documentation) the operational range can be modified to suit the specific requirements of the mill.

4. Maintenance

- The spray tubes are to be regularly inspected (at least every 4 weeks) for damage and to see that the outlet holes are free. For this purpose they can be externally withdrawn from their supports. If this is not possible then they must be inspected and repaired if necessary from inside the mill. Caking of the mill feed on the spray pipes must be removed as far as possible. If the spray pipe has to be removed it can be cut off at the end of the support and the rest can be externally withdrawn.
- Water hoses are to be protected from physical damage and exchanged if necessary.
- Couplings are to be newly sealed with tape or hemp after repairs have been carried out.
- The pipeholders on the mill body are to be regularly checked for security.

Seal Air Monitoring cpl.

Table of Contents:

	Sheet
1. General	1
2. Construction and Function	1

1. General

Sealing air is used in the roller and rocker arm assemblies (see the corresponding chapters) to prevent the ingress of dust into the bearings. The sealing air monitor causes an alarm to be activated in the event that the sealing air supply is disturbed.

2. Construction and Function

The sealing air stream is monitored by a electronic pressure transducer at each roller. A ball valve situated between the air stream and the sensor allows the sensor to be exchanged whilst the mill is in operation. Under normal circumstances the valve must be kept in the open position.



Replacement of the pressure transducer only with the valve closed!

LM Drive cpl.

The LOESCHE mill drive is substantially composed of

- the mill motor (in certain cases supplied by Customer) with motor base frame,
- the mill gear and
- the coupling between mill motor and mill gear with
- appropriate coupling guard.

The mill motor will drive the dish via the flexible coupling and the mill gear. The dish is directly bolted to the gear output flange.

The mill motor is supported by the motor frame which will be screwed and pinned to the motor base frame embedded in concrete after alignment of mill motor in relation to mill gear.

The coupling guard ensures that no accidental contact of rotating parts will be possible during grinding.

In case of damage and after dismantling of mill motor (complete with motor frame) and lifting of dish the mill gear may be withdrawn for hauloff from underneath the mill on so-called track rails welded onto the motor base frame embedded in concrete.

For further details as to installation, operation and maintenance of mill gear please refer to the gearbox's separate documentation.

LM-Platform cpl.

1. Construction

The mill platform is arranged around the mill at the same level as the top of the LM-lower-housing. The bridges that connect the stands serve in the longer axis of the mill as platforms, too. The bridges have a teardrop or chequered plate top surface to make them non-skid.

Big mills have also platforms at the stands. For maintenance work at the rollers there also can be a removable service platform.

In front of the stands there are parts of the platform that are attached to the bridge-sides. They can be dismantled for the swinging-out of the rollers.

Access to the platform is possible from a building platform or from a stairway.

2. Servicing

During swinging-out process of the rocker arm/roller unit the platform parts in front of the bearing blocks and at the stands must be removed.



During dismantling of the platform and the following maintenance work on the mill safety belts are to be used to prevent accidents through falls.

LM-Auxiliaries: General

Special tools and auxiliaries which are necessary for maintenance the machines belong to the delivery extent of the mill.

Along with this a list containing the requested standard tools for assembly and maintenance is supplied. The standard tools have to be supplied by the customer.

LM-Auxiliaries: Table

Assembling/ Dismantling of Table

For assembling and dismantling of the mill gearbox the table must be loosened from the gearbox cover flange, lifted and supported.

Dismantling

- The cover, if fitted, is to be removed from the center of the table.
- The bolts of the table have to be removed.
- The hydraulic pressing-off cylinder(s), which belong to the auxiliary tools, are now bolted with their flanges to the table after insertion of distance pieces.
- Drive out the hydraulic pressing-off cylinders with the hand pumps supplied. Ensure that the table is evenly lifted!
- This procedure is to be repeated until the table has reached the distance from the gearbox cover flange that is shown in the drawing.
- The drive pins have to be clear of the bores of the table.
- In this position the table must be put on its supports.
- After supporting the table, the hydraulic pressing-off devices can be removed.



For safety reasons, the condition of the supports is to be checked before usage.

After removal of the gearbox, it is not permitted to enter the area below the propped-up table.

The area must be cordoned-off!

Assembling

- When remounting the table the procedure takes place in the opposite manner.
- Please make sure that the corresponding bores of the table for the drive pins in the gearbox cover flange are positioned above the drive pins.
- Lower the table without tipping onto the gearbox cover flange.
- The bolts of the table must be tightened according to the notes in the assembly drawing "Table cpl." and finally secured.

LM-Auxiliaries: Roller

Installation/Dismantling of Roller

Dismantling

For dismantling the roller, i.e. to extract it from the rocker arm lifting eye nuts and/or eye bolts are provided. Details can be taken from the illustration of the drawing "Auxiliaries: Roller".

The pressing off cylinder is to be bolted against the face of the rocker arm. With the help of the hand operated pump of the table dismantling aid the pressing off cylinder looses the conical sleeve on the outside of the rocker arm and presses the roller axle further out of the second conical sleeve unit in the rocker arm. Then the roller can be extracted from the arm by means of a crane.

Installation

Make sure when assembling a new roller that the conical clamping surfaces of the two clamping sleeve units are free of dirt and fretting corrosion. The conical surfaces must be bright. They have to be treated with a thin film of bearing lubricant installation to prevent fretting corrosion. Then the roller is tightened in the position which is indicated on the drawing "Roller cpl.".

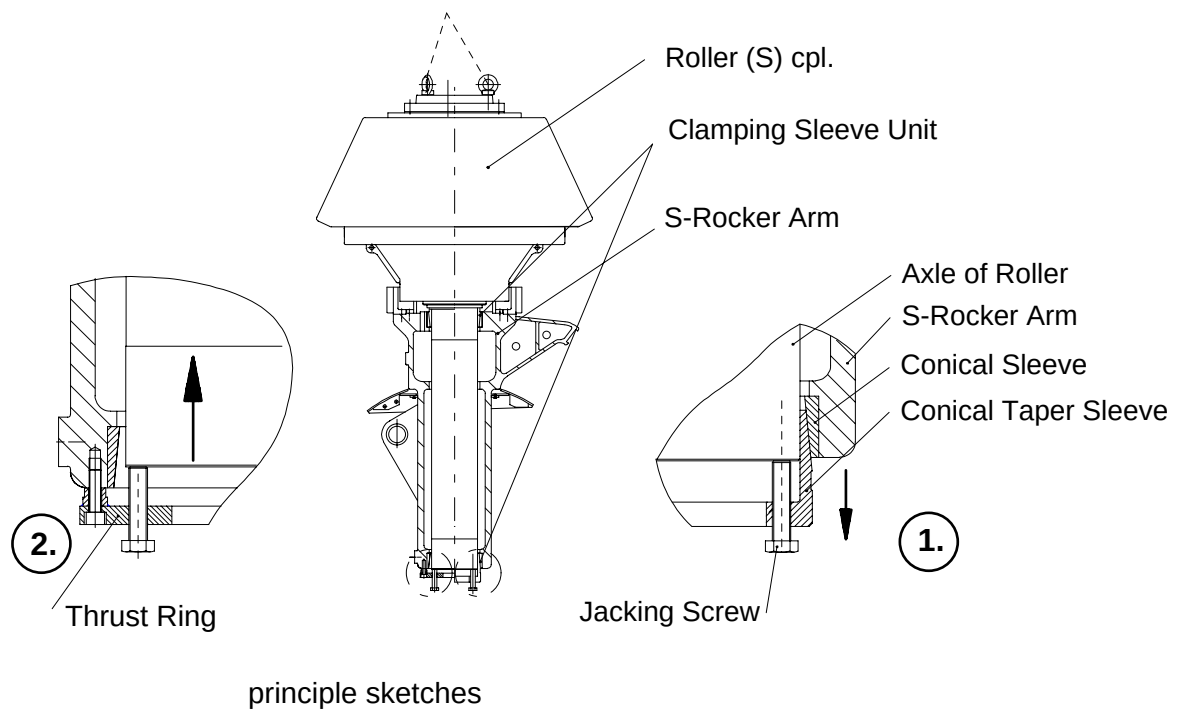
Please observe:

- Position of the slot on the face of the roller axle must be in the correct position according to the drawing "Roller cpl.".
- The slit in the taper sleeve must be plugged tightly with lead according to the drawing "Rocker Arm cpl.".

If a temperature sensor exists, see operating instructions "Temperature Sensor for Roller".

LM-Auxiliaries: S-Roller

Installation/Dismantling of S-Roller



Dismantling

To assist the removal of the roller, jacking screws are provided for the taper sleeve inside the rocker arm.

After removal of the fastening bolts between the flanged taper sleeve and the roller axle the jacking screws may be fitted. These jacking screws serve to pull out the taper sleeve from the rocker arm. Following this, the thrust ring can be fitted, which serves as a reaction point in order to force the second conical joint apart. The roller can be then suspended from previously fitted eyebolts in order to be lifted out of the rocker arm by means of a crane.

Installation

Make sure when assembling a new roller that the conical clamping surfaces of the two clamping sleeve units are free of dirt and fretting corrosion. The conical surfaces must be bright. They have to be treated with a thin film of bearing lubricant installation to prevent fretting corrosion. Then the roller is tightened in the position which is indicated on the drawing "Roller (S) cpl.".

Please observe:

- Position of the slot on the face of the roller axle must be in the correct position according to the drawing "Roller (S) cpl.".
- The slit in the taper sleeve must be plugged tightly with lead according to the drawing "Rocker Arm (S) cpl.".

LM-Auxiliaries: Roller

Dismantling of the Taper Roller Bearing

A so-called hydraulic unit has been provided to facilitate removal of the taper roller bearing from the roller. The hand-lever pump used for the purpose of dismantling the table and roller, serves for the injection of highly viscous oil into the grooves of the roller bearing seat. The injected oil expands the bearing seat at the core for easier dismantling of the bearing.

The dismantling process necessitates the connection of a hose to the corresponding borehole in the hub which is usually sealed by a threaded pin. (refer to drawing "Roller cpl.").

Dismantling of the Cylinder Roller Bearing

This requires the screwing of bolts into a number of boreholes on the roller core, to push the cylinder roller bearing from the bearing seat.

LM-Auxiliaries: Roller

Dismantling of Roller-Rolling Bearings

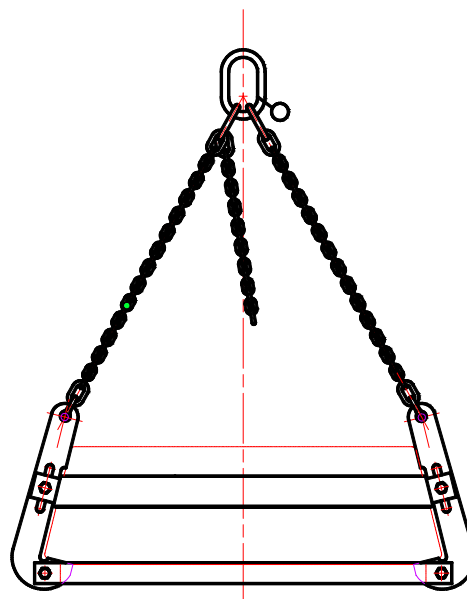
To facilitate the dismantling of roller bearings from the roller so-called hydraulic bandages are provided. By means of the hand pump, which is used for the dismantling of the table and the roller, highly viscous oil is pumped into the grooves of the roller bearing seatings. The oil expands the roller seating in the hub to facilitate the dismantling of the bearings.

For dismantling of the bearings a hose must be connected to the respective bore in the roller core which is usually sealed by a threaded plug (see drawing "Roller cpl.").

LM-Auxiliaries: Roller

Lifter cpl.

With lifter and hoisting gear the roller tyre is lifted for mounting or dismounting. This lifter must not be used to loosen roller tyres from the roller hubs by means of hoisting gear. It is laid out only to be strong enough to carry the weight of a new roller tyre.



Lifter

Principle Sketch

Change of roller tyre

- Swung out roller and the rocker arm out of the mill
- Loose the nuts of the hammer bolts
- Remove the hammer bolts and the clamping ring of the roller
- Loose the roller tyre from the roller hub!
- Therefore heat the roller tyre all around evenly with a gentle flame

Check the condition of the lifter before usage.



The lifter may only be used for the transport of roller tyres. Removal by force from the hub is not permitted.

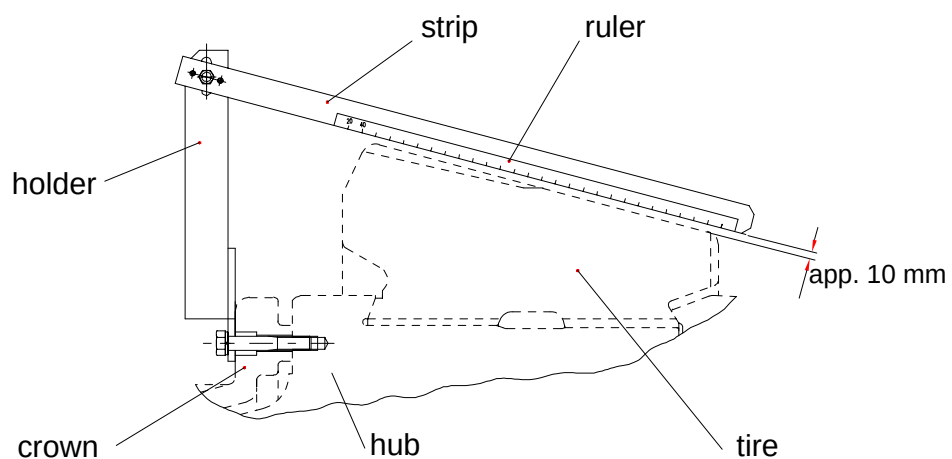
Pay attention to the maximum permissible load which the lifter may carry. Danger of accidents! (see drawing "Lifter cpl.")

LM-Auxiliaries: Roller

Tire Profile Ruler cpl.

The wear of the grinding parts should be checked regularly in order to accumulate important data regarding the projected life expectancy and to permit comparisons in case of changes in grinding stock and grade of materials of grinding parts.

Important for meaningful measurements is the comparability. LOESCHE has provided the means for such measurements with the tire profile ruler.



principle sketch

Because this measuring device is designed for various sizes of rollers, the exact adjustment and fixing can be made only when the unit has been fitted to the roller, according to the drawing "Tire Profile Ruler cpl.".

The ruler should be initially adjusted to have a gap of approx. 10 mm to the outer surface of the roller tire. This is necessary since new grinding rollers may vary slightly on account of the rough surface of the cast tire. The size of this gap in new condition must be noted and all subsequent measurements must have this value subtracted from them. For the evaluation of the measured values we recommend the use of a table corresponding to our drawing 3.F 1158.

LM-Auxiliaries: Roller

Oil Drain Device cpl.

During the swinging-out movement of the roller, the oil, intended for the lubrication of the roller bearings, returns through the central bore of the roller axle. The swinging-out procedure should be slow enough to give the oil sufficient time, even at low temperatures and high viscosity, to flow to the end face of the roller axle.

Prior to swinging-out of the roller, the oil-wetted filter has to be removed and the oil drain device has to be assembled. Thereto an adapter and a gate valve are screwed to the axle of the roller. In accordance with the drawing "Oil Drain Device for Roller" a crystal-clear hose with clamp is attached to the slide. The hose permits the observation of the return flow of the oil. The hose is to be placed into a receptacle of adequate capacity.

Before commissioning the mill it should be ascertained that the drained oil is returned or replaced with clean oil.

LM Auxiliaries: Rocker Arm

Assembly/Dismantling of the Rocker Arms

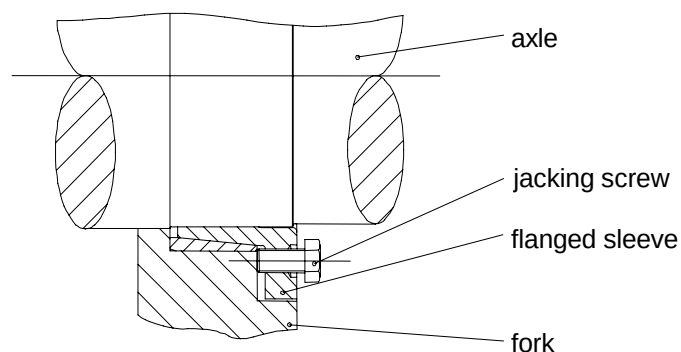
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1. Flanged Sleeve	1
2. Taper pin connection	2

Before swinging the rocker arm/roller unit out of the mill, the flanged sleeve which connects the fork with the axle (refer to "Rocker arm cpl." assembly) must be loosened, as well as removing completely the taper pin unit which connects the lever and the fork.

See also construction drawing "Rocker Arm cpl.".

1. Flanged Sleeve



Principle sketch

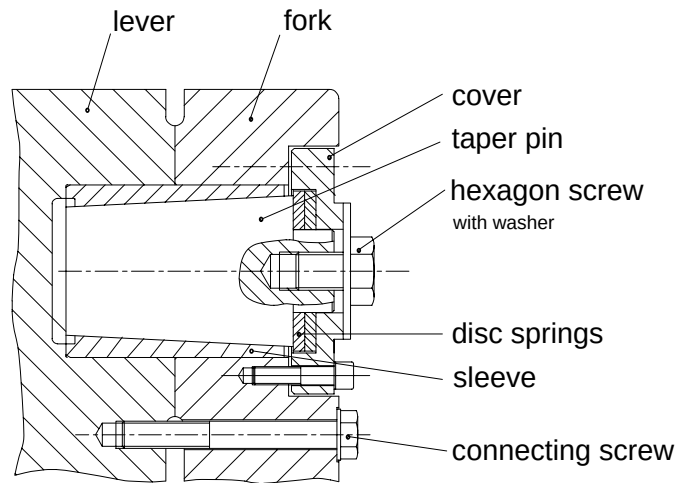
Dismantling

- Loosen clamping bolts,
- Tighten jacking screws until flanged sleeve is forced out of contact with its mating part,
- Pull the flanged sleeve out until it lies adjacent to the bearing block.

Assembly

- Remove the jacking screws,
- Clean and treat the seating surfaces with rust preventive,
- Using the clamping bolts, pull the flanged sleeve into engagement and tighten the bolts.

2. Taper pin connection



principle sketch

Dismantling

The cover should be carefully released, and the disc springs located beneath it removed. Once the cover has been resecured, a hexagon screw is screwed into the centre of the taper pin. The effect of this is that the taper pin will move outwards into the rearward recess of the cover, being released from its mounting in the sleeve in the process, and relieving the tension on the slotted sleeve element. When the cover is released, the connecting components can be removed.



The hydraulic system must be without pressure before dismantling.

The attachment bolts may only be removed after loosening the taper pin connection.

Assembly

The gap between the front face of the lever and the fork must be adjusted to the dimension shown in the "Rocker arm complete" drawing.

The slotted sleeve is pressed into the superimposed borehole of the fork and hub, and then tensioned by the taper pin, which is driven in with the aid of disc springs, cover and screws. This causes the sleeve to engage with a positive fit against the borehole wall of the lever and fork.

Before carrying out reassembly, please ensure absolute cleanliness and make use of assembly lubricant.



Before clamping the sleeve with the taper pin, the connecting screws of the lever and fork are to be checked for tightness.

The disc springs are to be tightened to a block.

LM Auxiliaries: Hydropneumatic spring system

Assembly/Dismantling of pin, bearings and spring cylinder

- a))The pin and spherical roller (or plain) bearing of the rod head situated at the lower end of the fork and
- b) The pin in the lower part of the pedestal (see assembly "Mill Stand Cpl.") for the bearing in the base eye of the cylinder
- c) The spring cylinder from its base eye

Please note the warning symbol and commentary to be found in the drawing "Auxiliaries: Spring Assy. Cpl."



The described operations may only be executed when the hydraulic system pressure has been reduced to zero.

Dismantling

a) Rocker arm/rod head

Rod heads equipped with spherical plain bearings

After disconnecting the rod head from the cylinder by removing the clamping nut it can be pulled off of the pin using the disc, spindle and collar nut (pos. 3 - 5). The rod head must first be supported using the eyebolt. The cover on the outside of the rod head is removed and the specified parts are attached to the rod head and pin as shown in the drawing.

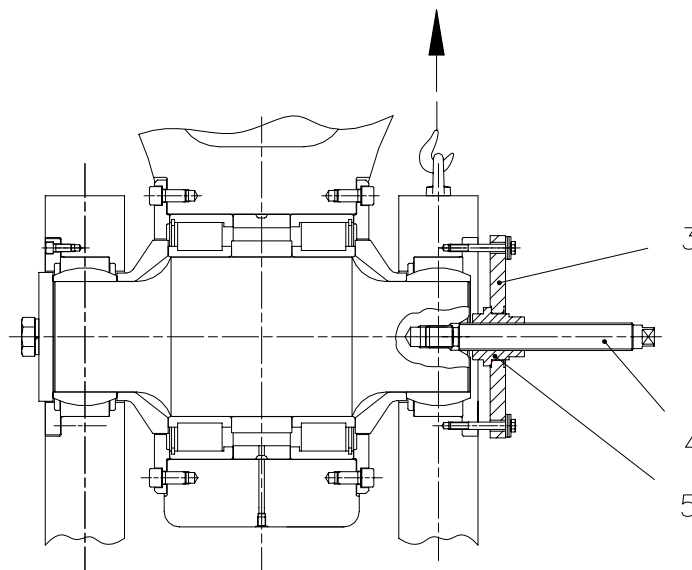


Fig. 1: Pulling off the rod head

Pin

As a first step, the mounting device pos. 1 is attached to one side of the fork and the disc, spindle and collar nut (pos. 3 - 5) are mounted onto it. On the opposite side of the fork mounting device pos. 2 is attached to the pin. This transfers the thrust from the pin to the outer race of the bearing. In this manner the pin, complete with both bearings and the intermediate ring, is pulled into mounting device pos. 1. Since the available length of the spindle is too short, the second step involves inserting a pipe, pos. 8, between the disc and pos. 1 in order to completely extract the unit from the fork. Finally mounting device pos. 2 must be removed before hanging pos. 1 from a crane by the eyebolts in order to lower the pin to the ground.

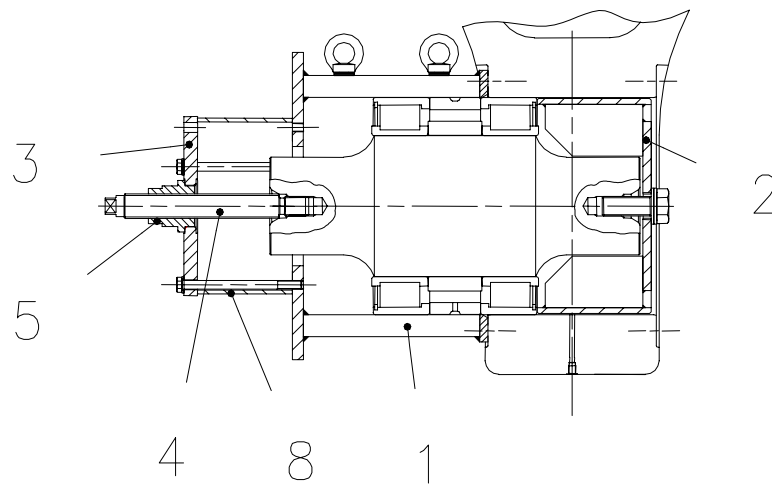


Fig. 2: Pulling out the pin

b) Base eye bearing of the spring assembly cylinder

The pin can be pressed out of its seating in the pedestal with the aid of the spindle, disc and collar nut (pos. 3 - 5). If fretting corrosion should mean that the spindle alone is not strong enough for this task, three small jacks can be attached to the inner collar of the intermediate ring to assist the removal. These jacks are not a part of our supply. On the outer side of the base eye a two-piece pipe, pos. 10, is attached with a strap around the pin. When the pin has moved out of its seats in the pedestal, further movement strips the spherical roller bearing and the base eye from the pin. This operation can be helped by the application of high pressure oil from the handpump in the assembly "Table Cpl." to the oil groove in the pin.

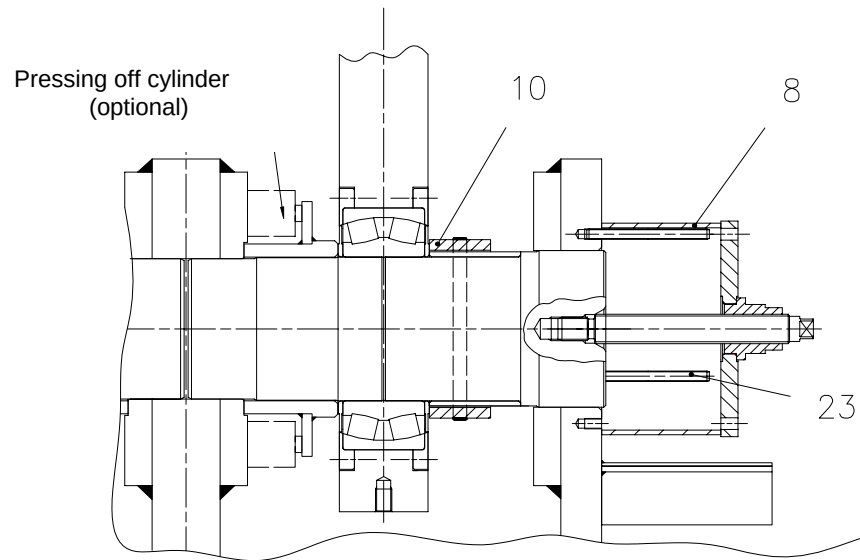


Fig 3 Extraction of the pin in the base eye

Assembly

Generally speaking, all components must be thoroughly cleaned and treated with preservative before their reinstallation in the machine.

a) Rocker arm/Rod head

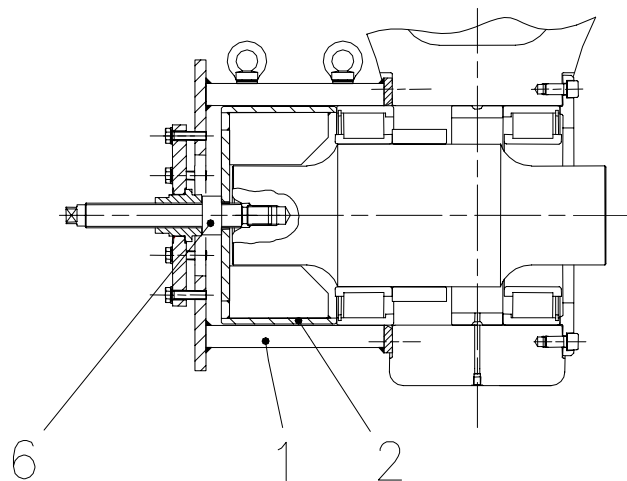


Abb.4: Installation of the pin into the rocker arm

Firstly one bearing with its cover and the outer intermediate ring are to be installed in the fork. The other bearing with the inner intermediate ring are mounted on the pin. The pin with mounting device pos.2 is then inserted into mounting device pos.1. Now the spindle is screwed into the pin and the nut, pos. 6, is turned until it lies against the mounting device pos. 2. Thereafter the collar nut and the disc can be attached. The entire unit can now be lifted up to the fork, taking care that the pin does not tilt and slip out of mounting device pos. 1. The pin is guided so far into the fork that the mounting device pos. 1 can be screwed to it. By turning the collar nut the spindle presses the pin and the bearing into the fork.

The rod heads can be refitted with the help of the spindle similarly to the arrangement shown in fig. 1. In this case, however, the collar nut must be installed in front of the disc.

b) Base eye bearing of the spring assembly cylinder

The assembly of the pin as well as the dismantling thereof is accomplished with the aid of the spindle, disc and collar nut as shown in fig. 5

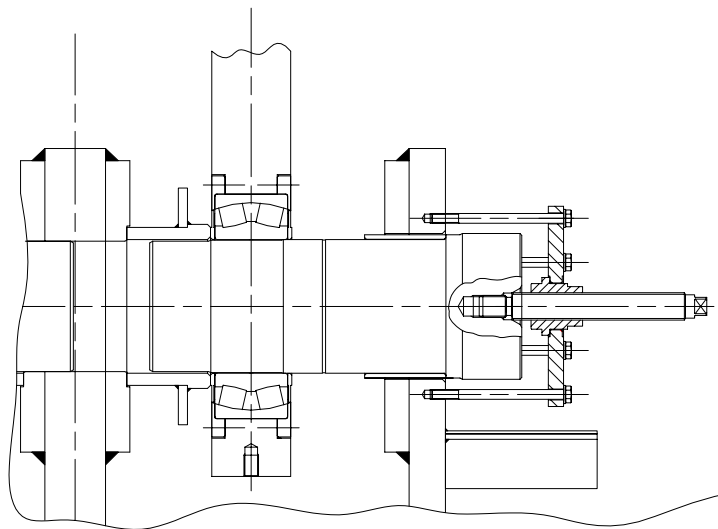


Fig. 5: Installation of the pin into the base eye

c) The spring cylinder from its base eye

It is not necessary to remove the base eye bearing assembly in order to remove a cylinder. The torque multiplier and socket which are part of our supply can be used to remove the bolts which connect the cylinder to the base eye. Thus the hydraulic part of the cylinder can be removed from the pedestal without disturbing the base eye. When reinstalling the cylinder it is imperative that the bolts are tightened to the torque given in the suppliers documentation.

LM-Auxiliaries: Hydropneum. Spring Assembly System

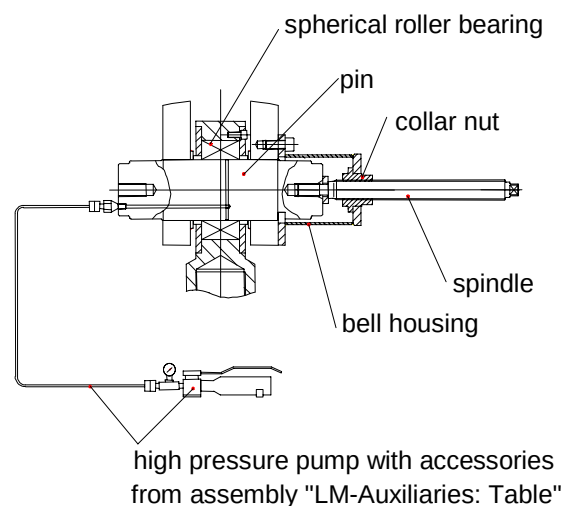
Assembly/Dismantling the Spring Pin of the S Spring Assembly

The pins for connecting the hydraulic cylinders of the S spring assembly to the S rocker arms and the covers of the LM upper housing, are easily fitted and removed with the help of the auxiliaries.

The spindle used as push and pull rod should be greased prior to use, in order to guarantee its smooth operation.

Dismantling

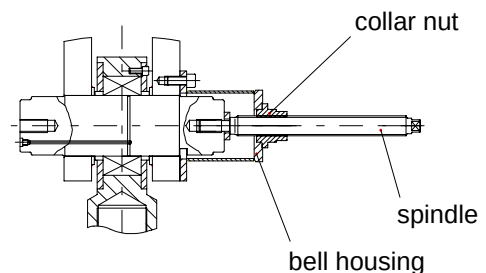
In the event of damage to the rolling bearing and dismantling of the hydraulic cylinder, the pins have to be removed from the respective bores. The seat of the spherical roller bearing on the pin may be widened hydraulically by means of the oil-hydraulic process. For that purpose, the disks with the screws fitted on the face of the pin, are removed and the hand-lever operated high-pressure pump included in the supply is connected to one end of the pin via a hose. The pin is then pushed from the boreholes by means of the device also part of the supply.



Assembly

The same device is used for re-assembly of the pin, the collar nut being inserted the other way round in that case.

Principle Sketches



LM-Auxiliaries: Hydropneum. Spring Assembly System

Filling and Testing Device for Bladder Accumulators and Piston Accumulators

This device is a combined appliance for filling the bladder and piston accumulators with nitrogen and a measuring instrument. By means of the filling and testing device it is possible to measure the gas pressure in the hydraulic accumulators. Filling with gas as well as measuring the gas pressure may only be accomplished when the oil pressure is zero. In this connection (see operating instructions of the manufacturer).



As gas only nitrogen is to be used. The use of compressed air or oxygen is strictly forbidden. **Danger to life!**

The bladder accumulator filling pressure must be chosen in the right relation to the hydraulic working pressure. The allowed limits are to be taken from the "Diagram: Spring Stiffness" (see parts list of documentation "HSLM").